

Supplementary Information

Spatial aggregation alters population-weighted estimates of InSAR observation support

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Overview and experiment definitions

The Supplementary Information has seven parts: A, domains and provenance; B, allocation mechanism and numerical checks; C, missingness and availability; D, sampling and processing; E, population benchmarks; F, regional transfer; and G, additional figures. The table below summarizes the domains, inputs and denominators used in these analyses. Population-model allocations are distinguished from Census counts and survey estimates throughout. Support deficits describe observation support and do not classify deformation.

Design/domain and period	Population and normalization	Support/target	Denominator and role
Chao Phraya finite source footprint; 2016–2022 pairs	GHSL 2020 native allocation; crossed WorldPop uses common coverage and GHSL total	Mean area-pair unwrapped/coherence support; allocated support deficit	Fixed 22.501-million GHSL allocation; crossed analyses have their own documented common total. Mechanism and input sensitivity.
Bangkok artificial-mask rectangle; fixed model snapshots	GHSL population/built surface; same coarse totals for all rules	Removed model allocation and constructed threshold classes	Original test-domain total; also known removed allocation for relative error. Controlled estimation.
Bangkok LiCS-BAS rectangle; 2019–2020	GHSL on the processing grid; originally final-valid population 11.281 million for LOS masking	437-pair input support, final binary mask, or hidden fitted-LOS class; these are separate targets	Full rectangle for quality/bounds; common-computable originally valid allocation for conditional mask error.
Fresno rectangle and complete blocks; 2020 population	GHSL native for footprint; Census uniform or within-block GHSL/CA-POP normalization for block tests	DWR published observation-cell union, without inferring rates in gaps	The denominator is the rectangle total or the complete-block total, as specified for each analysis.
Fresno irregular-block masking; 2020-10-01–2021-10-01	523,953 Census residents; 23 lack usable GHSL weight; 523,930 normalized units	Hidden fine-pixel class under random/contiguous masks; adapted whole-unit median rule	522,596.78 units in known-rate footprint; 1,333.22 outside it. Common-computable denominator for error; full known domain for unestimated share/bounds.
Fresno ACS/residential benchmarks; ACS 2017–2021, Census 2020, municipal vintage in S27	ACS point/MOE and 1-km grid totals; held-out calibration uses training groups only. Residential rules normalize within eligible blocks	Administrative total/allocation agreement; eligibility-constrained support sensitivity	319 complete groups, ACS total 493,571. Residential joint domain: 466,999 or 467,490 Census units.
Fresno DWR rate transfer; exact intervals in S28	Fixed 2020 Census block counts assigned to each block’s median annualized vertical class	Whole-block median-rate assignment; distinct from hidden fine-pixel class prediction	501,411 residents in common rate-covered blocks, from 523,953 in all complete blocks. Temporal sensitivity.

Design/domain and period	Population and normalization	Support/target	Denominator and role
Los Angeles and Davis-Sacramento; archived ARIA track periods in S29/S30	GHSL allocation; fixed regional totals. GNSS stations have separate metrics	Finite nonzero LOS plus coherence screening; external LOS agreement under two frames	Regional total for allocation; disjoint validation stations for GNSS. Transfer magnitude and limits.

“Native” means retaining the declared population model on its analysis grid. In controlled-mask comparisons, prevalence estimates hidden class allocation from the class fraction in the retained population allocation. “Allocation discrepancy” compares fixed models/rules; “estimation error” uses known artificially hidden values; “internal displacement residual” compares excluded interferograms; “external LOS agreement” compares the independent GNSS observations with the archived external product. Percentage points (pp) always refer to the stated denominator.

Part A: Domains, provenance and published practice

S1. Frozen domains and source provenance

Experiment domains and parameter settings are recorded in `config.json`. The primary domain consists of GHSL 3-arc-second pixels whose centers fall in the frozen finite footprint of the archived Chao Phraya VLM cells. Pixel rows in experiment NPZ files increase from south to north; GDAL raster rows and LiCSBAS binary files increase from north to south. Scripts explicitly convert between these conventions. No velocity-unit assumption enters the primary domain.

Table S1: Main input products and the information they provide.

Product	Version/selection	Role and interpretation
Chao Phraya VLM	Zenodo 18092985, v2; public 1-km GeoTIFF	Frozen finite geographic footprint; WabInSAR source. Archived $\times 10$ screen is supplementary and not independently established by raster unit tags.
LiCSAR	Frame 062D_07629_131313; 28 legacy pairs and 224 expanded pairs, 2016–2022	Pairwise coherence and valid-unwrapped support diagnostics, not WabInSAR validity.
LiCSAR matched stack	Both dates in 2019–2020, baseline ≤ 96 days; 437 catalogue pairs	Separate small-area LiCSBAS processing and date-selected edge holdout.
GHSL	R2023A, GHS-POP and GHS-BUILT-S, 2020, 3 arcsec; 2015/2020, 30 arcsec	Extensive population and built-surface allocation models. Pixel scale is not independent demographic precision.
WorldPop	Thailand 2020, unconstrained UN-adjusted, 3 arcsec, record 29165	Alternative count distribution. Common-total results isolate placement differences conditional on common validity.
ESA WorldCover	2021 v200, tile N12E099, 10 m	Fractional cover strata within GHSL pixels; not a 2020 occupancy map.
DWR	Vertical Displacement Point Data Locations 2026Q1, layer 0	The service contains measurement-cell polygons. We calculate their union before allocating population.

Source file URLs, hashes, geometries and download records are retained under `external/`, `results/fine/` and `results/lineage/`. The DWR fixed rectangle is selected before support results are computed. Its 38,738 polygons have 38,738 unique source codes. Their clipped-area sum and union agree to about $7 \times 10^{-5} \text{ m}^2$ (floating-point roundoff), confirming that repeated source cells do not inflate assigned population. The union area is 388.905 km² within a 495.510-km² domain.

S2. Original binary statistic and sensitivity analysis

The submitted cell table and its archived staging copy have the same SHA-256, `96ea01a6233315e8c37f06c55a5f0dac3656ae70239fded6b90aa68a63982a8a`. Both reproduce 12,589 selected legacy cells, population 3,772,492.0508 and built surface 404.772254 km². The alternate closure branch has hash `a17161efe405c98d01a221a549ee9e33a8cc3950a290d36c9a1f0b71ced68429` and yields zero cells under the same tabulated selection rule. The submitted table reproduces the reported total; the alternative archive branch contains a different selection.

The legacy total sums whole-cell population selected by a center-majority rule. Point registration, unwrapped validity, support-statistic choice and population allocation change different parts of that calculation; they are not additive corrections to a validated physical exposure count. The revised mean area-pair deficit uses an explicitly different estimand.

The following factorial comparison holds archived cell centers and exposure weights fixed. The “raw byte” cutoff of 0.3 is intentionally retained only to diagnose incorrect encoding. GDAL point registration and added unwrapped validity are changed independently. Table S2 concerns the legacy center-majority statistic; the primary analysis uses the mean area-pair deficit on the geographic domain.

Table S2: Binary center-majority lineage audit under the inherited strong screen. Population is in millions of full-cell allocation units.

Coherence encoding	Registration	Joint validity	Selected cells	Population
Raw byte (incorrect)	Legacy tiepoint	No	0	0.000000
Raw byte (incorrect)	Legacy tiepoint	Yes	822	0.133146
Raw byte (incorrect)	GDAL point	No	0	0.000000
Raw byte (incorrect)	GDAL point	Yes	822	0.133819
Byte / 255	Legacy tiepoint	No	12,589	3.772492
Byte / 255	Legacy tiepoint	Yes	12,644	3.834586
Byte / 255	GDAL point	No	12,582	3.756674
Byte / 255	GDAL point	Yes	12,607	3.797393

The full CSV additionally reports continuous center-frequency deficits for each combination and both domains. A center-majority rule, mean within-pixel area-pair support, and the fraction of area reaching a specified pass-count threshold are different statistics. For example, spatially uniform frequency 0.5 and a half-area split between frequencies 0 and 1 have the same mean, but different fractions meeting a frequency threshold of 0.5. Threshold distributions in this study refer to mean support at the GHSL grid, not an unobserved infinitely fine persistence distribution.

The VLM file has SHA-256 `5393e5c865c5105705b4b0e915f9133591816933965c88b67dfd273db6f2603d`. It contains no explicit stored-velocity unit tag. In the 18,077 frozen cells, directly applying a numerical cutoff of -5 to raw values selects no cells; applying it after the inherited multiplication by ten selects 17,808. This sensitivity explains why the primary experiments use a geographic domain. The archived metadata do not establish the factor-of-ten conversion. We therefore restrict this conversion to the documented sensitivity analysis. The newer version 18093069 of the source repository contains random-forest code; it does not supply an explicit replacement unit record for this GeoTIFF.

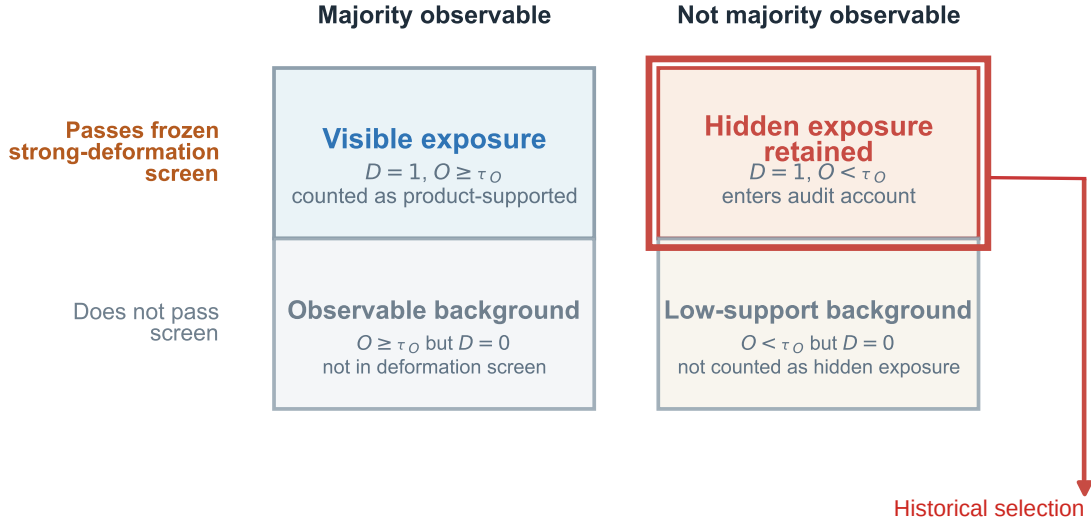


Figure S1: Historical binary cell classification. Here D denotes membership in the inherited deformation-screen class, O is the product-support fraction and τ_o is the majority-observable threshold. The labels *hidden exposure* and *majority observable* are historical labels, not the primary estimand or a verified exposure claim. The original threshold selection is audited in Section S2. Continuous support and allocation summaries in main Sections 2.1 and 3.1–3.3 and main Figs. 2–3 define the primary analysis.

S3. Downstream exposure examples and the Iran provenance correction

Table S3: Targeted examples of downstream exposure practice and the scope supported by each study.

Source	Reported measurement and population-transfer choices	Relevance and comparison limits
Blackwell et al. [1]	California coastal VLM combines ALOS/Sentinel-1 and GNSS; community exposure is estimated assuming uniform population distribution within communities.	A community-scale transfer assumption materially defines the exposure estimate; the authors describe it as a guide for prioritizing detailed risk assessment, not a complete risk inventory.
Fernández-Torres et al. [2]	Mexico-wide assessment applies a temporal-coherence threshold of 0.7, compares InSAR velocities with GPS, and estimates population/households in detected subsiding urban localities using census information.	The published exposure is tied to the mapped localities and screened observations; it does not by itself estimate deformation in areas without usable InSAR observations.
Ao et al. [3]	Reports urban land and population shares above specified InSAR subsidence-rate thresholds across 82 major Chinese cities.	A population share associated with mapped urban subsidence classes; it should not be interpreted as a counterfactual estimate for unobserved city areas.
Ohenhen et al. [4]	28 US cities mapped with WabInSAR at about 28-m resolution; rates were compared with 154 GNSS stations. Citywide summaries use area-weighted rates; 2020 census-block population is assigned to velocity classes using the median InSAR-pixel rate in each block.	A high-resolution, geodetically checked exposure workflow and a valuable alternative. Area-weighted city summaries and census-block median-rate assignments answer different questions from fine-grid population-weighted support.

Source	Reported measurement and population-transfer choices	Relevance and comparison limits
Murray et al. [5]	Masks unreliable InSAR observations and combines retained VLM estimates with a lidar DEM for future coastal flood exposure. The discussion notes that subsidence between large buildings may remain undetected.	The study identifies undetected motion between buildings as a coverage limitation and calls for ground surveys or higher-resolution observations.
Aieorattanawadee [6]	Bangkok Sentinel-1 time series for 2018–2022 compared ascending and descending results with seven CORS stations; the thesis abstract reports agreement at four and two stations, respectively.	A local example of independent geodetic comparison, but sparse station coverage. The published abstract does not provide station-level series for reanalysis here, so it is not validation of our results.
Luachapichatikul [7]	2016–2019 descending Sentinel-1 PS versus leveling; leveling/PS relative rates (mm yr^{-1}) -3.32/-4.28, -7.08/-6.49, -2.62/-3.93 and -1.84/-0.88; PS-to-benchmark offsets 15.32, 196.59, 120.34 and 34.89 m; leveling-fit R^2 0.82, 0.94, 0.86 and 0.10.	Three rate differences are 0.59–1.31 mm yr^{-1} , but PS is descending LOS while leveling is vertical; offsets are material, one fit is poor, and the sample is four selected pairs. This is validation practice context, not validation of our stack.
Pumpuang and Aobpaet [8]	Bang Kapi PS-InSAR compared with RTSD SBKK CORS over October 2019–February 2020; reports $r = 0.88$ and RMSE 3.98 mm. InSAR at the station was unavailable, so the comparison used mean change within a 100-m buffer.	Directly relevant independent-validation practice, but one station and a buffered spatial target. The published summary is not raw validation data for our analysis.
Keerinart et al. [9]	A 23-week SBKK CORS analysis (2 October 2019–6 March 2020) reports a vertical ellipsoidal-height rate of 13.308 mm yr^{-1} ; the authors note construction and pile driving at the station building during the observations.	A nearby-in-time geodetic record, but the authors identify a site disturbance that can make monument motion differ from regional ground motion. Its plotted samples are not machine-readable station data.
Haghshenas Haghighi and Motagh [10]	Nationwide Sentinel-1 analysis of 2014–2020 observations; public annual-rate raster projected from LOS to vertical, a derived subsidence mask, and seasonal-amplitude raster.	The archived rate and mask are from this 2024 source, not a Motagh et al. (2008) nationwide rate product. These summaries do not provide the per-pair support field used in the primary experiment.
Payne et al. [11, 12]	2014–2022 analysis of 106 selected subsiding regions; ascending/descending data are decomposed into vertical and east–west velocities with 1-sigma uncertainties. Regions are screened by rate/area and interpretation criteria.	A newer and relevant scientific product, but its released component-velocity fields and selected-region masks are not pair-specific support masks. Its underlying LiCSAR interferograms are separately accessible; this study does not reprocess them for Iran because Iran is no longer a quantitative application domain.
LiCSBAS [13]	Unwrapped-data checks, loop closure, inversion and multi-index quality masking.	Established processing quality; a sampled coherence diagnostic alone is insufficient.

The Bangkok examples in Table S3 document local geodetic validation practice. Their published summaries do not provide the complete station series and processing metadata required for an independent matched reanalysis of our stack.

SBKK support check in the separate LiCSAR stack. The published RTSD coordinate for SBKK ($13.792944^\circ \text{ N}$, $100.596494^\circ \text{ E}$) lies inside the matched LiCSAR subregion [14]. At this coordinate, the nearest LiCSAR grid-cell center is 44.8 m away and passes the final LiCSBAS mask. Two pixel centers lie within a 100-m radius, and both pass the mask; their median velocity is 2.314 mm yr^{-1} in LOS. This is a product-specific support audit, not GNSS validation. It differs from the cited PS-InSAR study because the sensor product and processing chain differ. We do not compare the LOS rate directly with the published SBKK

ellipsoidal-height rate: the motion components and reference frames differ, and the SBKK paper reports construction and pile driving during the GNSS interval. The accompanying `SBKK_Support_Audit.zip` contains the lossless local filtered-data window, all 104 acquisition dates, geometry and parent-file hashes. Running `reproduce_sbkk_support.py` reproduces `reference_results/SBKK_100m_LiCSAR.csv` and `reference_results/SBKK_support_summary.json` without requiring the full processing stack. The portable replay and the original full-stack script give byte-identical outputs.

The archived Iran rate and mask match the official Zenodo 10815578 files by MD5 (respectively `c0a19300c161940dc2e6a8de267cef4f` and `734c58f4f2f2fbc7017d705f9b81fe3d`). This establishes the archive provenance only; neither raster enters an exposure result. The archived seasonal-amplitude file does not match that release and is also unused. Although Payne et al. make regional time-series velocities available through the COMET-LiCS Subsidence Portal and the source interferograms through the LiCSAR portal, Iranian datasets are excluded from the quantitative analyses in this study. The exploratory Iran parser used the 2024 product; its earlier attribution to a 2008 nationwide product was incorrect. The file-level audit is retained in `results/lineage/iran_actual_file_match.json`.

These examples compare reporting units and support definitions; they were not selected to estimate how common a reporting practice is. We excluded the earlier EGMS point-based totals because counting an entire population pixel at every radar point double-counts population. LOS measurements also require a geometric decomposition before they can be interpreted as vertical displacement.

Part B: Allocation mechanism, numerical checks and reporting scale

S4. Geometry, conservation and numerical checks

Coherence and unwrapped validity are intersected on the union of their actual raster boundaries before integration. A disjoint-mask analytical test confirms that their intersection is zero even if both separately averaged masks are one-half. Further tests cover extensive mass splitting, partial target coverage, half-pixel offsets, valid zero coherence and spherical area conservation.

All 512 production scale/origin/exposure/domain/method combinations preserve the native reference and the appropriate extensive total within the stated numerical tolerance. In the baseline summaries, the largest relative error in $S_E + U_E = T_E$ is 1.99×10^{-15} . The largest absolute covariance-identity residual is 5.96×10^{-8} in the corresponding exposure units. Positive-, negative- and zero-covariance analytical examples verify the sign convention independently of the raster implementation. The exact empirical weighted-distribution integral matches the direct population deficit at printed precision.

The UTM transfer uses GDAL sum resampling of extensive first and cross moments. The GDAL sum reprojection approximates the transfer of extensive quantities; exact spherical polygon intersections are evaluated separately. Production 50-m transfers close their totals within 10^{-8} relative tolerance. Recomputing from native inputs at 25 m produces measured relative total drifts of -1.66×10^{-5} for area, -2.06×10^{-5} for supported area, -4.11×10^{-6} for population and -3.28×10^{-6} for supported population. These diagnostic fields are not renormalized. Their maximum combined effect on the 1-km uniform population estimate is 2.40×10^{-6} of the AOI population, or 0.00024 percentage points. The diagnostic drift is reported separately from the production conservation result.

These checks assess numerical conservation and geometric consistency. Population and displacement accuracy require separate comparisons with independent observations. The numerical-check record is included in `Reproducibility_Records.zip` as `supporting_records/numerical_checks.json`. The conservation and convergence tables are released as `revision_2026/reference_outputs/core/allocation/conservation.csv` and `integration_convergence.csv` in the same directory. Section S5 gives the complete path index.

S5. Reproducibility files

The fixed release v0.6.3 contains the current submission documents and portable audits alongside the scientific reference outputs retained from v0.6.0. Selected printed macro values are recorded in `revision_2026/manuscript_number_ledger.json`; this file is not a complete mapping of all results to CSV tables. The additional experiment records are in `revision_2026/additional/scientific_number_ledger.json`. The file `revision_2026/submission_20261003_final/section_output_index.json` maps current Sections S1–S31, historical script identifiers and local result paths to the released reference tables and audit packages.

The accompanying `Reproducibility_Records.zip` includes this index, the numerical-check summary and the published Bangkok comparison table with its source audit. Index paths are relative to the repository

root; the reference tables themselves are obtained from the fixed release. The two standalone audit packages are `SBKK_Support_Audit.zip` (S3) and `Local_Validation_Audit.zip` (S17). Each contains its own inputs, provenance and replay instructions.

In the table below, paths beginning with `core/` are relative to `revision_2026/reference_outputs/`; paths beginning with `supporting_records/` are relative to `revision_2026/submission_20261003_final/` and are also included in `Reproducibility_Records.zip`.

Analysis	Released result locations
Original binary statistic	<code>core/lineage/archive_branches.csv</code> ; factorial center-rule audit
Native support summaries	<code>core/fine/allocation_summary.csv</code> , <code>core/fine/support_curves.csv</code>
Scale and origin effects	<code>core/allocation/scale_origin_comparison.csv</code>
Missingness experiments	<code>core/masking/replicates.csv</code> , <code>core/masking/synthetic_hazard_bounds.csv</code>
Sampling dependence	<code>core/sampling/</code> ; date network and retained rank plans
Population and cover	<code>core/population/</code> ; own and common coverage
Second-region geometry	<code>core/dwr/</code> ; polygon-union audit and masking
Matched processing	<code>core/timeseries/</code> ; stage and holdout records
Published comparisons	<code>supporting_records/literature/bangkok_leveling_pairs.csv</code> and <code>bangkok_leveling_source_audit.json</code> in the same directory.

S6. Method choices and comparison scope

Table S4: Relationship to existing methods and the evidence supplied by this study.

Approach	Required information and interpretation	Evaluation here
Center sampling	Coarse population totals and support at a representative center; subcell support is assumed representative.	Same-input scale/origin comparisons; boundary effects separated; controlled-mask variability retained.
Uniform areal allocation	Coarse totals and support-area fractions; population assumed uniform over included area [15].	Main method contrast, local errors, covariance decomposition, opposite-direction and small-bias scenarios.
Native-model allocation	Fine population model and intersected support; uniformity remains within each source population pixel.	Controlled reference with conserved first and cross moments; retains model detail without household validation.
Built-area weighting	Coarse totals and fine ancillary building surface, consistent with dasymetric allocation principles [16].	Tested under introduced masks; shared information with GHSL limits independence.
Observed-domain overlay	Population assigned only to a stated valid observation footprint.	Fresno uses actual DWR cell polygons and their nonduplicated union; no inferred motion in gaps.
Processing quality diagnostics	Interferogram network, inversion outputs and multiple quality criteria [13, 17].	Full/train LiCSBAS branches and excluded-edge residuals; not equated with the primary sampled-support statistic.
Imputation or additional sensors	An explicit deformation model or additional independent observations with compatible references.	Discussed as information needed for missing deformation; no numerical superiority over these different-target methods is claimed.

Uniform allocation is adequate for the chosen model target when within-cell population–support covariance is negligible at the required reporting scale. Native allocation retains that covariance when fine input information is available. These are conditional choices; a method that retains inaccurate fine population structure is not thereby demographically validated. Complete-case summaries remain valid for their explicitly observed domain. We use these established allocation rules to quantify differences under matched inputs and identify the conditions under which those differences are small.

S7. Boundary sensitivity and numerical verification

For the primary $q = 0.3$ population field, an additional implementation constructs the geographic domain from the union of native GHSL mask pixels. Edges are densified at native-pixel spacing before transformation to UTM. Coarse squares fully covered by this boundary form the complete-interior subset; all remaining included squares form the partial-boundary subset. Different origins produce different interior subsets, so each row uses its own population and reference deficit. We independently recalculated the projected moments using separately constructed interval weights; all 32 primary method/scale/origin estimates agree with the retained table within 10^{-4} population units.

Table S5: One-kilometre uniform allocation after excluding boundary-crossing squares. Deficits are in million mean population-allocation units. Relative difference uses the interior reference; population share uses the original whole-domain total.

Origin shift (m)	Cells	Pop. share (%)	Native	Uniform	Difference (%)
(0, 0)	14,120	99.49	2.960	4.685	58.29
(500, 0)	14,107	99.39	2.950	4.668	58.24
(0, 500)	14,123	99.41	2.955	4.685	58.56
(500, 500)	14,111	99.27	2.944	4.664	58.46

The interior contrast remains 58.24–58.56%, compared with 57.7–57.9% across the full domain. The separate center-outside subset quantifies the explicitly declared zero-support boundary rule in the center method. Full per-origin tables retain cell counts, denominators, signed errors, absolute cell-error sums and positive/negative cell counts for both methods. Files are in `revision_round2/evidence/boundary_allocation.csv`.

For native support, a separate nested-rectangle intersection calculation checked 80 deterministic random/support-extreme pixels for every original pair and all four support definitions (8,960 comparisons). The largest support-fraction difference was 2.98×10^{-8} , consistent with float32 storage. Actual GeoTIFF tiepoints, pixel scales and point registration were independently compared with GDAL transforms for both raster types. These are sampled spatial checks, not exhaustive independent reprocessing of every native pixel.

The internal recheck reconstructed all 44 held-out edge residuals directly from retained phases and the training HDF5 series, verified 393 training memberships, and recalculated the training-only support bins. The largest difference in grouped RMSE was 7.27×10^{-7} mm under independent float64 arithmetic, compared with reported precision of 0.001 mm. Each evaluable pixel–edge observation receives equal weight; none of these counts is assumed to be an independent physical replicate.

Each region’s complete 21,600-row controlled-mask factorial was checked for coverage and arithmetic. Repetitions 0, 1, 49 and 99 were regenerated for all mechanisms and removal fractions at the 10-pixel zero-origin setting, independently recalculating 144 method results per region. All 400 conditional sampling draws and all 64 population-model/coverage/support summary rows were recalculated. Nested mean fields are stored as float32, so averaging already-integrated pair totals and integrating a stored mean can differ slightly; the maximum observed difference was 1.35×10^{-8} of the corresponding total exposure.

For Fresno, an independent global union of the original polygon rings differs from the stored covered-area sum by 0.0019 m^2 over approximately 388.9 km^2 ; 128 deterministic random pixel intersections agree within 2.34×10^{-11} in support fraction. All 1,260 distinct original radar rasters referenced by the legacy, expanded and matched manifests match their recorded checksums.

S8. Covariance identity and scale sensitivity diagnostic

Define the normalized deficit $D = U/T$, where U is deficit allocation and T the fixed total allocation. The expressions below are dimensionless shares; multiplying by 100 gives percentage points, not absolute population units.

For a reporting square containing fine pixels i , let A_i be pixel area, $s_i \in [0, 1]$ the observed-support fraction and $\rho_i = p_i/A_i$ the native population density. With the same total population and geographic square held fixed, the difference between the area-uniform and native-population support deficits is

$$D_{\text{uniform}} - D_{\text{native}} = \frac{\text{Cov}_A(\rho, s)}{\bar{\rho}_A}, \quad (1)$$

where area-weighted moments are calculated over the square and $\bar{\rho}_A = \sum_i A_i \rho_i / \sum_i A_i$. The absolute

difference is bounded by $\sigma_{\rho,A}\sigma_{s,A}/\bar{\rho}_A$ by Cauchy–Schwarz. This identity describes the accounting mechanism; it is not an empirical uncertainty interval.

We checked this identity on a fixed common domain of 120,175 pixels, crossing the GHSL and common-total-normalized WorldPop surfaces with the 437-pair mean input-support field and the final full-stack quality mask. Projected pixel centers were assigned to 250-, 500-, 1,000- and 2,000-m squares at four grid origins. The resulting scale/origin ranges are:

Table S6: Uniform-minus-native deficit and Cauchy–Schwarz upper bound across projected square sizes and grid origins. Values are percentage points. Pixel-center grouping is a sensitivity diagnostic and differs from the direct-intersection Bangkok paired-support estimates in S19.

Support field	Population surface	Difference range	Bound range
437-pair mean	GHSL	0.550–4.670	1.076–9.599
437-pair mean	WorldPop, normalized	0.164–2.376	0.429–5.320
Final quality mask	GHSL	0.012–0.241	0.088–1.967
Final quality mask	WorldPop, normalized	0.010–0.197	0.054–1.200

The covariance identity residual is at most 1.22×10^{-15} percentage points across all 64 model–support–scale–origin combinations, and no observed absolute difference exceeds its corresponding bound. The grouped-square differences are not pooled with the direct-intersection Bangkok paired-support estimates in S19; their spread across scales and origins is design sensitivity, not a confidence interval. Population-model assumptions are described in S22. Input checksums, full scenario definitions, local effects and machine-readable results accompany the analysis outputs.

S9. A tolerance-based reporting rule for scale-dependent allocation error

The bound requires the declared fine-scale population and support inputs, or externally justified bounds on their within-cell variation. Coarse population totals alone cannot supply it. For a reporting application with a prespecified acceptable aggregate error ϵ , the analytic Cauchy–Schwarz upper bound can be compared with ϵ for each population model, scale and grid origin. A bound at or below ϵ certifies that the allocation discrepancy cannot exceed that tolerance under the stated inputs. A bound above ϵ is inconclusive; it does not demonstrate that the observed error is unacceptable. The tolerances below (0.5, 1 and 2 percentage points of total allocation) are illustrative tolerances with no external reporting-standard basis.

The following numerical illustration uses the Fresno common Census-block domain and the two 100-m GHSL and CA-POP allocation surfaces, each normalized within blocks to the same Census totals, with the DWR polygon-union support field defined in S25. The denominator is 523,908 residents in 5,713 common blocks. At 250 m, all four origins for both GHSL and CA-POP have bounds below 1 pp (maximum 0.58 and 0.88 pp, respectively). At 500 m, none of the eight model-origin combinations meets 1 pp; at 2 pp, all four GHSL origins pass and no CA-POP origin passes. At 1 km and 2 km, neither model meets 2 pp for every origin. The bounds rise with scale, while local L1 error and net error can differ substantially and top-decile spatial overlap declines. Thus a small net aggregate difference alone need not preserve local ranking. The detailed 24 model-scale-tolerance summaries and source 32-row audit are in `reporting_tolerance/tolerance_summary.csv` and `scale_method_sensitivity.csv`; these are deterministic finite-domain diagnostics, not sampling uncertainty.

S10. Spatial decomposition of Bangkok support-allocation effects

To show whether the AOI-wide allocation effect hides local reversals or changes in low-support labels, we grouped the matched Bangkok processing grid into projected 1-km and 5-km squares aligned to integer-kilometre coordinates in UTM Zone 47N. Within each square, the native estimate weights the pixel support by the GHSL allocation; the area-uniform estimate applies the square’s area-weighted support fraction to the same square population total. The reported difference is uniform minus native missing-support population, divided by the fixed AOI allocation total. Pixel-center grouping is used here as a local diagnostic and is kept separate from the direct-intersection Bangkok paired-support estimates in S19.

At 1 km, the difference is 3.194 percentage points for the mean support across 437 input pairs and 0.103 points for the final quality mask. At 5 km it is 6.580 and 0.389 points, respectively, illustrating scale dependence. The 5-km tile contributions for the final mask include both signs and cancel 13.2% of their

summed absolute effect; at 1 km the cancellation fraction is 39.8%. For a descriptive 50% support cutoff, native-population and area-uniform unit labels differ in 150 1-km units containing 139,903 GHSL allocation units for the mean input-support field, and in 17 units containing 2,229 allocation units for the final mask. These are units whose labels disagree, not a net change in population assigned to the low-support category. The 50% cutoff is neither a hazard threshold nor a universal operational rule.

Table S7: Local decomposition of support-allocation effects in the fixed Bangkok AOI. Deficits are percentages of the common GHSL allocation total; signed differences are percentage points. Sign-cancellation fraction is calculated across all reporting squares at the listed scale. Discordant population is allocation mass in squares whose native and area-uniform support fractions fall on opposite sides of 50%.

Support field	Scale (km)	Native deficit (%)	Uniform deficit (%)	Difference (pp)	Sign cancellation (%)	Discordant units / GHSL units (count / allocation)
437-pair mean	1	3.454	6.648	3.194	1.9	150 / 139,903
437-pair mean	5	3.454	10.034	6.580	0.0	12 / 123,735
Final quality mask	1	0.294	0.398	0.103	39.8	17 / 2,229
Final quality mask	5	0.294	0.684	0.389	13.2	0 / 0

The final-mask effect at the primary 1-km scale is small, and the number of threshold-discordant allocation units is also small. The larger mean-input-support effect should not be interpreted as final-product failure: the two support fields summarize different stages of the processing chain. Tile-level results, input checksums and the calculation script are provided in the analysis bundle.

Part C: Missingness, estimator availability and class bounds

S11. Artificial masking, error denominators and synthetic bounds

Each region has 21,600 method/scale/origin outputs: four missingness mechanisms, three removal fractions, 100 repetitions, three coarse sizes, two origins and three allocation rules. The same mask is used for all methods within a repetition. Gaussian smoothing generates spatial masks; density-associated scores combine standardized log density with a spatial term. The fixed seed is 20260927. Removal fractions refer to available fine pixels, while the realized removed area fraction is saved separately. In Fresno the controlled reference domain is the original published observed footprint, including its fractional intersections with GHSL pixels.

Table S8: Bangkok masking at 30% removal, 10-pixel cells and zero origin. Error is relative to known removed allocation. Ranges are conditional mask percentiles.

Mechanism	Method	Mean error (%)	2.5th–97.5th (%)
density negative	area uniform	120.35	[105.55, 134.61]
density negative	building weighted	19.60	[17.46, 22.28]
density negative	cell center	102.97	[86.96, 121.20]
density positive	area uniform	-14.17	[-15.15, -13.03]
density positive	building weighted	-4.99	[-5.52, -4.47]
density positive	cell center	-11.64	[-13.98, -9.39]
random	area uniform	0.04	[-0.40, 0.44]
random	building weighted	0.03	[-0.24, 0.27]
random	cell center	0.08	[-9.19, 8.76]
spatial	area uniform	0.01	[-0.82, 0.81]
spatial	building weighted	0.01	[-0.50, 0.50]
spatial	cell center	0.01	[-3.60, 3.12]

When the removed population is small, relative errors can be large: low-density-associated removal at 10% of pixels is such a case. Figure S17 normalizes error by the entire test-domain population, so a large ratio is not mistaken for a large fraction of all residents. Raw reference totals, estimates, signed errors, cell RMSE and absolute cell-error totals remain available for every scenario. Building-weighted improvement is conditional on GHSL’s shared built-environment information and should not be generalized to independently observed household locations.

The synthetic hazard fields are experiment definitions, not inferred or measured deformation. With threshold membership $h_p \in \{0, 1\}$ and introduced missingness m_p , the omitted threshold allocation $\sum_p E_p h_p m_p$ lies between zero and $\sum_p E_p m_p$. Adding observed threshold allocation gives the identification interval in the main

text. All synthetic checks satisfy this deterministic bound. Complete outputs are under `results/masking/` and `results/dwr/masking/`.

The Fresno repeat shows the same sign reversal with different magnitudes (Fig. S2). At 30% removal, 10-pixel cells and zero origin, uniform-area mean error is -16.14 percentage points of the observed-footprint population for high-density-associated removal and $+7.28$ points for low-density-associated removal. The corresponding built-weighted errors are -10.61 and $+4.23$ points. Uniform-area mean errors for random and unrelated spatial masks are -0.021 and -0.0025 points. Relative-to-removed-population errors can be very large in sparsely populated masks; both denominators are retained, and the figure uses the whole observed-footprint total. The direction of the allocation error depends on the association between missingness and population density.

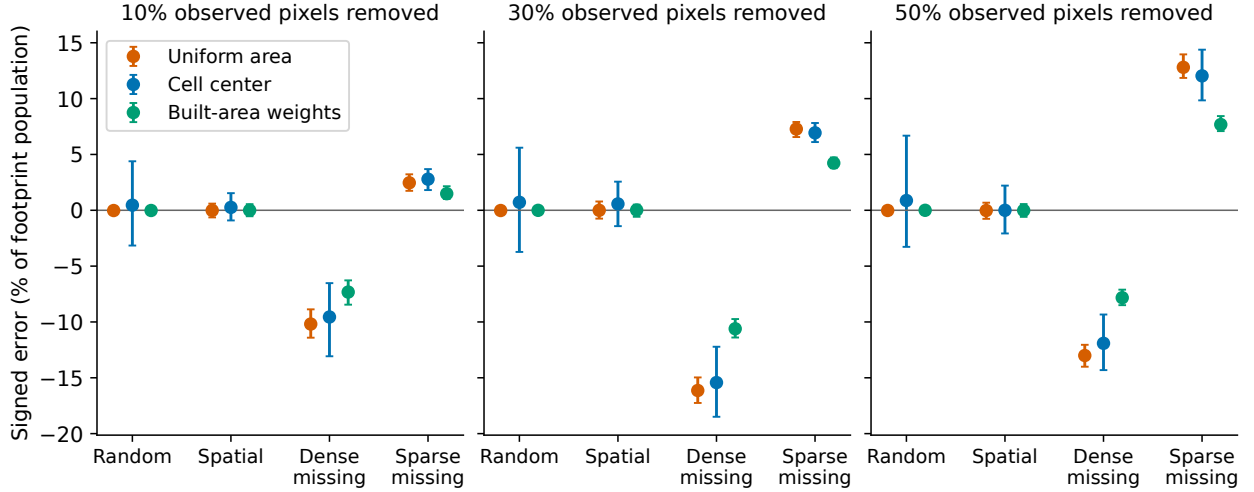


Figure S2: Controlled masking within the original Fresno observation footprint. “Dense” and “sparse” indicate preferential removal from higher- and lower-density fine locations. Identical masks are supplied to all three methods. Symbols and bars summarize 100 repetitions by their means and central 95% mask ranges, conditional on the fixed GHSL reference. The denominator is population originally allocated inside the published footprint; the figure does not evaluate motion outside that footprint.

S12. Artificial masking of observed LOS classes

The reference is the unfiltered fitted line-of-sight (LOS) velocity from the full LiCSBAS branch, positive toward the satellite, restricted to its final validity mask. It contains 117,103 of 120,701 grid cells and 11.281 million GHSL allocation units. Population-weighted shares at or below -2 , -5 and -10 mm yr $^{-1}$ are 2.474%, 0.2805% and 0.0223%, respectively. These are relative LOS classes, not vertical-settlement or hazard thresholds.

For each of 100 seeded repetitions, we remove 10% or 30% of valid cells using random, spatially clustered, high-velocity-spread or low-velocity-spread scores. Quality-associated scores include a seeded random perturbation. We use 5-, 10- and 20-source-pixel square units (about 0.55, 1.11 and 2.22 km), four zero/half-cell origins and partial edge units. Each estimator predicts only the class allocation in the artificially hidden part of a unit. Its reference is the sum of the original fine-pixel class allocation there. The observed-population prevalence rule requires positive retained population. The center rule uses the nearest retained pixel; ties are averaged. It requires at least one retained pixel. A unit with no hidden allocation has the known hidden target zero; otherwise a rule without its required information remains undefined.

Errors are calculated on units for which every compared rule is defined. The common-domain denominator is their originally valid population allocation, including retained and artificially hidden portions. We also preserve each method’s own computable domain and unestimated allocation, normalized by the original full-domain total. Omitting undefined units from a conditional error calculation does not recover their class. Whole-domain bounds add all hidden allocation to the observed class allocation for the upper endpoint; they include hidden cells even where an estimator supplies a prediction.

For case r , the relative bias is $100(\hat{H}_r - H_r)/H_r$; cases with $H_r = 0$ are undefined for that ratio and are counted separately. The archive preserves both mean-of-ratios and the raw quantities needed to form a ratio-of-means. Percentile ranges elsewhere summarize seeded masks and origins; they are design sensitivities,

Table S9: Hidden-class estimation at 30% removal, 10-pixel units and $v_{LOS} \leq -5$ mm yr⁻¹. Reference and signed bias are means in GHSL allocation units on the common-computable domain. Mean relative bias is the mean of case-specific ratios, not the ratio of the two mean columns. The final column counts cases with a positive reference denominator. Availability and conditional signed/L1 errors are given in Table S11 for the overlapping design.

Removal	Rule	Mean reference	Mean bias	Mean relative (%)	Defined ratios
Random	Prevalence	9,514	+91	+1.1	400/400
Random	Center class	9,514	+9,455	+99.7	400/400
Spatial clusters	Prevalence	5,695	+3,999	+74.7	400/400
Spatial clusters	Center class	5,695	+7,970	+146.3	400/400
High spread	Prevalence	3,475	+8,164	+234.7	400/400
High spread	Center class	3,475	+11,090	+319.6	400/400
Low spread	Prevalence	509	+1,945	+378.7	400/400
Low spread	Center class	509	+2,420	+518.6	400/400

not confidence intervals. The common-domain prevalence bias is positive under high-spread removal after unavailable units are excluded. Neither prevalence nor a center class reliably recovers allocation under structured missingness. Velocity spread is a processing diagnostic, not a calibrated uncertainty interval.

The complete 57,600-row factorial and grouped summaries are in `actual_los_masking/`. S14 adds the median-class rule to the overlapping 1.11-km design. All these tests condition on the population model, originally valid domain and observed LOS values. Original invalid pixels supply no hidden reference.

S13. Finite-domain bounds for threshold-class allocation under the actual final mask

We calculated bounds on threshold-class allocation under the actual LiCSBAS full-stack final quality mask on the fixed 2020 GHSL allocation domain. Let T denote total allocated population units in the domain, U the allocated units in pixels unsupported by the final mask, and E_c the observed allocation in LOS class c among supported pixels. Without assumptions about the unsupported pixels' LOS class, the finite-domain class share lies in $[E_c/T, \min(T, E_c + U)/T]$. This is a worst-case accounting interval, not a confidence interval and not a statement that deformation occurs in unsupported pixels.

Table S10: Finite-domain allocation bounds for LOS threshold classes under the actual LiCSBAS final mask. Values are model allocation units and percentage points, not enumerated residents.

LOS threshold	Observed class units	Identified share interval (%)	Width (pp)
≤ -2 mm yr ⁻¹	279,088	[2.4666, 2.7610]	0.2944
≤ -5 mm yr ⁻¹	31,644	[0.2797, 0.5741]	0.2944
≤ -10 mm yr ⁻¹	2,516	[0.0222, 0.3167]	0.2944

The interval width is set by the unsupported allocation share (0.2944 percentage points). Conditional on the observed rates and fixed population allocation, both endpoints are attainable: assigning all unsupported pixels outside the class gives the lower endpoint; assigning them inside gives the upper endpoint, without changing any observed input. Thus the bound cannot be narrowed without additional information or assumptions about the missing class assignments. At -5 mm yr⁻¹ the upper share (0.5741%) exceeds twice the observed share (0.2797%); a small whole-domain width need not imply a small relative uncertainty for a rare class. Here U uses a binary final mask. The mean area-pair deficit is a different estimand and cannot replace it in this bound. These intervals do not propagate measurement or population-model error. Results were calculated by `partial_identification.py` from the retained full-quality raster and are in `partial_identification/bounds.csv`.

S14. Median-class assignment and hidden fine-pixel class allocation

We evaluated the observed unit-median rate rule under the 30%-removal, 10-pixel, -5 mm yr⁻¹ design in S12. There are 100 masks for each of four mechanisms and four fixed origins. A median-class indicator is multiplied by the hidden population allocation, and compared with the hidden fine-pixel class total. This

adapts the classification step of published Census-block median-rate workflows; it does not reproduce their distinct target of assigning the entire block population to its median class. The two targets can differ even without missingness (the unmasked irregular-block comparison in S15 quantifies that difference).

Prevalence requires positive observed population; median and center classification require an observed pixel. When the required retained observations are absent, the method returns no point estimate; the corresponding allocation is reported as unestimated. All three errors use the same intersection of computable units in each case, with their full originally valid allocation as denominator. Unit-level absolute errors are summed before normalization to give local L1. The unestimated share has the original full-domain denominator. Conditional error describes computable units, whereas the unestimated share records allocation for which the method provides no estimate.

Table S11: Estimator availability and hidden-class error over 400 mask–origin cases per mechanism. Unestimated allocation is a percentage of the original domain. Signed error, its 2.5th/97.5th percentiles and local L1 are percentage points of the common-computable domain.

Removal	Rule	Unestimated (%)	Mean signed	Lower	Upper	Mean L1
Random	Prevalence	0.0000	+0.0008	-0.0074	+0.0106	0.0392
Random	Median class	0.0000	+0.0190	+0.0007	+0.0399	0.1039
Random	Center class	0.0000	+0.0838	+0.0544	+0.1176	0.1479
Spatial clusters	Prevalence	11.3467	+0.0400	-0.0014	+0.1153	0.0900
Spatial clusters	Median class	11.3417	+0.0579	+0.0012	+0.1499	0.1182
Spatial clusters	Center class	11.3417	+0.0797	+0.0122	+0.1762	0.1321
High spread	Prevalence	18.1625	+0.0884	+0.0354	+0.1574	0.1071
High spread	Median class	18.1612	+0.1034	+0.0527	+0.2016	0.1271
High spread	Center class	18.1612	+0.1202	+0.0564	+0.2174	0.1424
Low spread	Prevalence	28.6576	+0.0242	+0.0067	+0.0503	0.0309
Low spread	Median class	28.6573	+0.0109	-0.0004	+0.0287	0.0202
Low spread	Center class	28.6573	+0.0301	+0.0105	+0.0741	0.0366

The prevalence rule has smaller mean local L1 than the median rule under random, clustered and high-spread removal; the median rule has smaller L1 under low-spread removal. The same ordering holds for absolute mean signed error here, but net error alone is not a measure of local accuracy. Under high-spread removal, all three common-domain mean signed errors are positive. The original-domain mean unestimated share rises from less than 0.0001% for random removal to 11.34–11.35% for clusters, 18.16% for high spread and 28.66% for low spread. Conditional errors cannot characterize these unavailable units.

Mean whole-domain bound widths are 30.003, 30.204, 24.122 and 34.904 percentage points for random, clustered, high-spread and low-spread removal. The widths depend on all hidden population, not only undefined predictions. The full 4,800-row table, method-specific denominators and bounds are in `published_method_benchmark/`. These artificial-mask results do not identify physical motion in genuinely unsupported locations.

S15. Actual irregular Census blocks and contiguous missingness

The reference domain contains 6,787 complete official 2020 Census blocks with P1 population 523,953. Blocks with no positive GHSL allocation weight contain 23 Census residents and cannot be normalized. Excluding those units leaves 523,930 normalized Census allocation units. Within each retained complete block, GHSL weights are normalized before selecting the known-rate footprint. This avoids forcing every Census allocation unit into observed locations. The known-rate footprint retains 522,596.78 allocation units, 38,569 published DWR cells and 65,108 exact block–cell intersections. The accounting is therefore $523,953 = 23 + 522,596.78 + 1,333.22$. Originally missing rates and the 1,333.22 units outside the known-rate footprint remain outside the controlled reference. The 23-person exclusion differs from the 45-person exclusion in S25 because that analysis uses a different raster grid/crop and common-support requirement; both are reported with their own denominator.

The selected DWR mosaic item is the annual interval 1 October 2020–1 October 2021, distinct from the older multi-year footprint. We retrieve raw floating-point values with no palette/rendering transformation and convert the provider’s feet to millimeters, divided by the exact interval duration in years. Rates are constant within published cells. Official NAD83 Census geometry and source-cell corners are projected to EPSG:3310 before exact intersections. Maximum polygon relative area-closure error is 2.31×10^{-10} ; the conservative GHSL transfer has relative mass drift 0.000233, after which complete-block normalization fixes

the Census total. The DWR footprint and rate layers share provider lineage; these are conditional method experiments, not independent geodetic truth.

With seed 20261001, each of 30 repetitions defines a random cell ordering and distances to one or four random spatial centers. The latter generate large contiguous holes on the irregular valid footprint. The same ordering nests 10, 30 and 50% removals. The three thresholds are -3 , -5 and -10 mm yr $^{-1}$. All 2,430 method-mask-threshold rows and nine unmasked baseline rows are retained.

Native observed-population prevalence extrapolates its measured class fraction to the block’s originally known allocation. Area prevalence uses the observed area fraction. The unit-median rule classifies a whole block using the median of retained published-rate cells whose centers fall inside that block. The median rule is undefined when a block contains no retained pixel center, even if a retained cell intersects the block. Population prevalence is undefined when retained population is zero; area prevalence is undefined when retained area is zero. Undefined units are not assigned zero. In the unmasked baseline, the median rule already has 412 undefined blocks representing 4.386% of known-footprint allocation. At -5 mm yr $^{-1}$, the native known class share is only 0.1308%. On the unmasked common-computable domain, median-class assignment has local L1 discrepancy 0.01167 pp, compared with zero for native prevalence by construction (area prevalence: 0.00561 pp). This baseline measures the difference between a whole-unit median-class rule and the fine-pixel class-allocation target; a lower discrepancy against the latter does not validate household locations or establish universal superiority over the published workflow.

We report both method-specific availability and signed/local-L1 errors on the intersection of computable blocks. Each repetition additionally retains method-specific conditional error and its exact denominator. On the entire originally known domain, class allocation is bounded by retained class mass and retained class mass plus all hidden allocation. Every known reference lies inside this accounting interval. Its width is hidden population allocation, not the removed pixel fraction, and is independent of the estimator. The numerical bound check is deterministic, not a confidence interval.

Table S12: Thirty-percent removal at the -5 mm yr $^{-1}$ cutoff: means over 30 fixed-mask repetitions. Unestimated share uses the known-footprint denominator; local L1 uses common-computable allocation. Bound width uses the whole known domain.

Mask	Method	Unestimated (%)	Local L1 (pp)	Bound width (pp)
Random	Native prevalence	0.20	0.00296	30.02
Random	Area prevalence	0.20	0.00608	30.02
Random	Unit median	9.41	0.01345	30.02
One hole	Native prevalence	23.01	0.00055	24.02
One hole	Area prevalence	23.01	0.00575	24.02
One hole	Unit median	26.80	0.01129	24.02
Four holes	Native prevalence	28.64	0.00197	30.97
Four holes	Area prevalence	28.64	0.00702	30.97
Four holes	Unit median	32.61	0.01332	30.97

At 30% removal, native prevalence’s unestimated allocation ranges from 0.064–0.307% for random pixels, 3.73–44.98% for one hole and 14.82–47.99% for four holes. Mean class bounds are [0.0903,30.1064], [0.0859,24.1010] and [0.0977,31.0655]%, respectively. Neither low conditional L1 error nor random-mask performance supports assigning a class to entire genuinely missing blocks. Full replicates, per-method denominators, bounds and source hashes are in `additional/real_block_masking/`.

Part D: Sampling dependence and processing quality

S16. Expanded sampling and temporal network

The catalogue snapshot contains 1,603 pairs with both dates in 2016–2022. The 224-pair pool comprises eight randomly ranked pairs from each of 28 year-quarter strata, alternating short (≤ 24 days) and long baselines as availability permits. An amendment swapped adjacent ranks in even quarters before support outcomes were extracted. Both the original rank plan and amended plan are retained. This improved baseline balance but did not force equal counts where long-baseline availability was limited.

The equal-pair mean describes this sample design, not a catalogue-frequency-weighted population parameter. Start-date season does not contain the full seasonal history of a long-baseline pair. The additional repeated samples draw 1, 2, 4 or 8 pairs per year-quarter without replacement from the fixed eight-pair stratum; their conditional distributions characterize design sensitivity. A 224-pair draw includes the entire

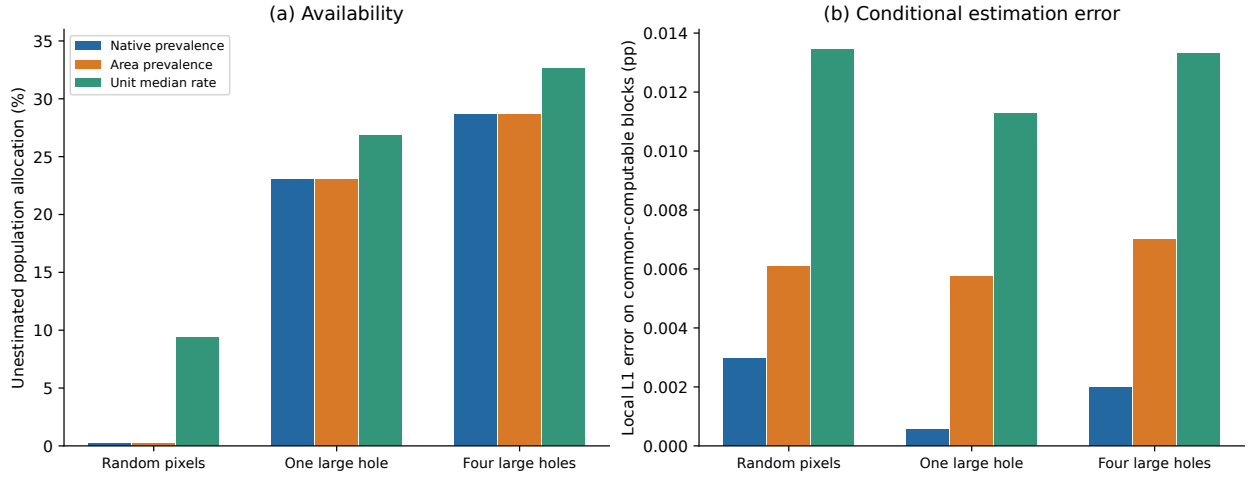


Figure S3: Availability and conditional estimation error on actual irregular Census blocks. Bars are repetition means for 30% removal and the -5 mm yr^{-1} cutoff. Each estimator uses identical missingness; the right panel omits noncomputable blocks by definition. The right-panel error is conditional on computable blocks; the left panel reports the allocation excluded because the estimator is undefined. Full 10/30/50% and threshold results are supplied in the CSV archive.

Table S13: Nested pool composition and population deficit at $q = 0.3$.

Pairs	Short baseline	Long baseline	Deficit (millions)
28	16	12	3.051
56	30	26	3.232
112	60	52	3.369
224	120	104	3.301

fixed pool and therefore has no resampling variation; variation outside this pool is not assessed. Shared acquisitions and atmospheric signals mean that individual pairs are not independent replicates of physical accuracy.

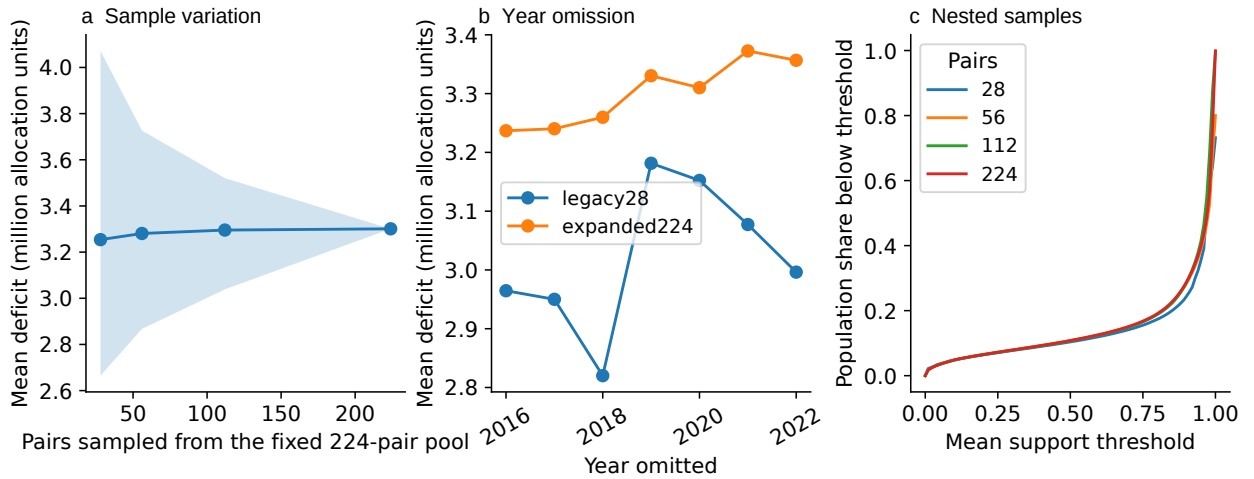


Figure S4: Conditional sampling variation, leave-one-year-out sensitivity and support distributions from nested samples. Bands in the first panel are the central 95% of introduced draws from the fixed pool; they are not confidence intervals for a complete time-series product.

Native pixels are aggregated by center into fixed 0.05° longitude–latitude blocks anchored at integer multiples of 0.05° . All blocks are retained in the CSV; rank comparisons use the 539 blocks with at least 1,000 GHSL population units and 5 km^2 inside the domain. Relative to the 224-pair pool, the nested 28-, 56- and 112-pair deficit-amount rank correlations are 0.9940, 0.9992 and 0.9994. Their top-decile Jaccard

Table S14: Date-graph diagnostics; these do not establish pixel-level connectivity.

Graph	Pairs	Dates	Components	Independent cycles
Legacy sample	28	54	26	0
Expanded pool	224	225	29	28
Catalogue 2016–2022	1603	288	1	1316

overlaps are 0.8000, 0.8621 and 0.9286. Deficit-share correlations exceed 0.9991. Thus similar overall ranks can coexist with changed totals and changed membership near a selection boundary. The 224-pair pool serves as a fixed comparator for sampling stability, not an independently validated reference. Spatial blocks are not treated as independent accuracy replicates.

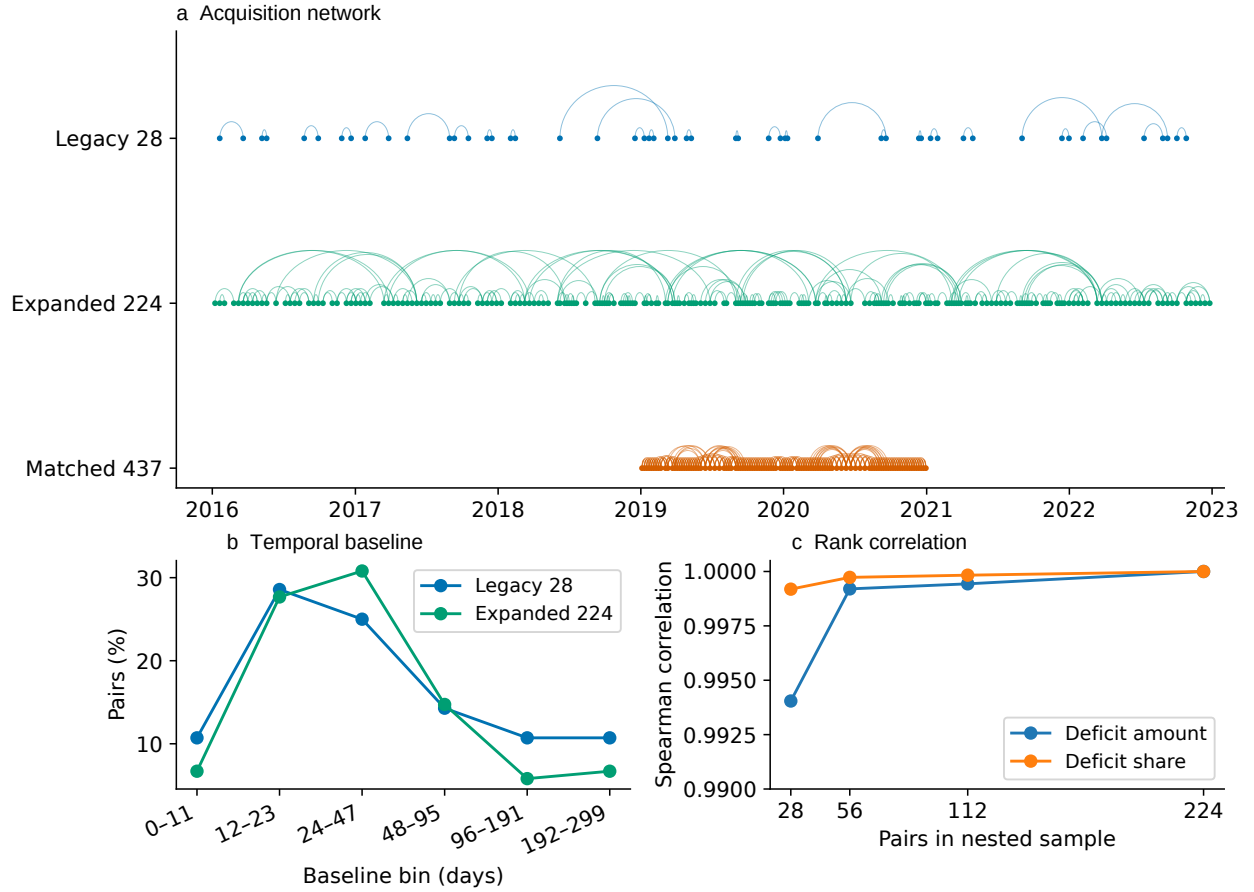


Figure S5: Actual acquisition-date graphs, temporal-baseline composition and descriptive block-rank sensitivity. Each arc joins the two dates of an input pair; graph connectivity is reported in Table S14. The matched graph precedes quality rejection. The rank panel uses a restricted vertical range to show small differences; all comparisons use the same 539 eligible blocks and the fixed 224-pair pool as comparator.

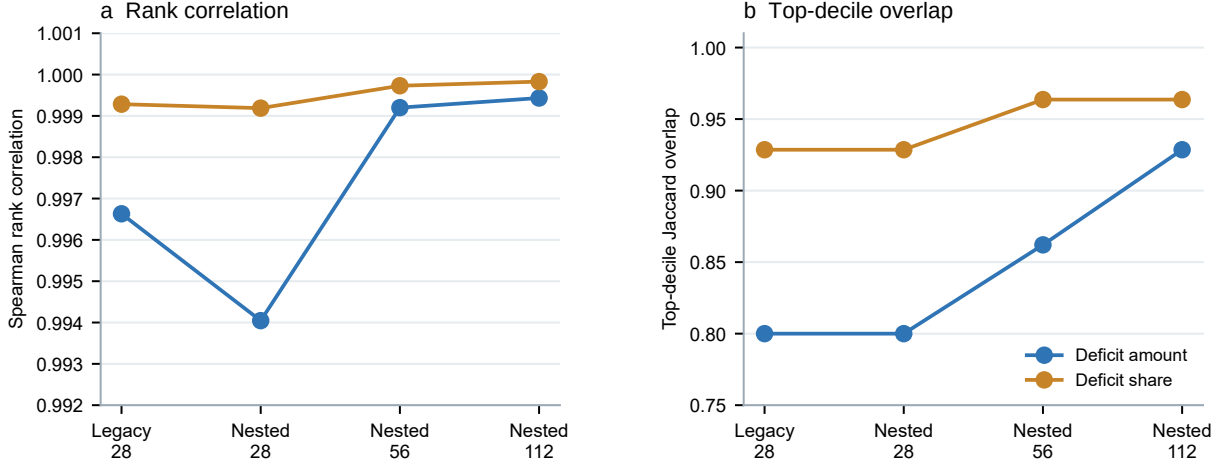


Figure S6: Descriptive ranking stability. All comparisons use 539 fixed geographic blocks, with nested 224 pairs as the reference. (a) Spearman correlation for deficit amount and deficit share. (b) Jaccard overlap of the 54 highest-ranked blocks under each statistic. Legacy and nested 28-pair selections are distinct. The restricted vertical axes show small differences. These ranks are not validated operational priorities.

S17. Matched LiCSBAS processing and holdout

We ran both branches through LiCSBAS2 steps 11–16 at commit 2627e50a1c703becd4a0b18754b05c1cf8c564d5, using the unmodified scientific routines. Platform-specific I/O changes are documented with the reproduction code. We prepared the inputs with Rasterio and ran the processing serially with a fixed NumPy seed and recorded package versions.

The processing raster has 301 columns and 401 rows, approximately 108 by 111 m, with its affine transform in `results/timeseries/geometry.json`. Frame 062D.07629.131313 supplies the inputs. Global network connectivity is required but does not ensure pixel-level connectivity. Pair checks retain 431 of 437 full inputs and 382 of 393 training inputs. Saved holdout/rejection lists document these selections; held-out phase/coherence never enters training.

Step 11 thresholds mean coherence at 0.05 and unwrapped coverage at 0.3; step 12 uses a 1.5-rad loop threshold. Step 13 applies least squares with regularization 0.0001 and requires more than 104 valid interferograms. Step 14 retains 100 upstream bootstrap draws and spatiotemporal consistency; bootstrap velocity spread is not calibrated physical uncertainty.

Step 15 uses fixed C-band defaults without automatic relaxation: mean coherence ≥ 0.05 , valid interferograms ≥ 156 , velocity spread $\leq 100 \text{ mm yr}^{-1}$, connected duration ≥ 1 year, temporal gaps ≤ 10 , spatiotemporal inconsistency $\leq 5 \text{ mm}$, interferograms without loops ≤ 50 , loop errors ≤ 5 and inversion residual RMS $\leq 2 \text{ mm}$. This mask shares coherence information with the $q = 0.3$ input diagnostic but adds other quality criteria. Mask logs use the upstream finite/non-reference denominator; our domain fractions include all 120,701 cells, including the finite reference pixel.

Step 16 applies the upstream 2-km spatial and 21-day temporal filter, without detrending, DEM-error fitting or height-linear correction. Both filtered products are retained; holdout prediction uses unfiltered training `cum.h5` to avoid filter-dependent smoothing. No atmospheric correction or external GNSS datum is added. Relative LOS is positive toward the satellite, using $-c/(4\pi f)$ with $f = 5.405 \text{ GHz}$, without vertical conversion.

For held-out pair (a, b) , prediction subtracts training cumulative displacement at a from that at b . Observed phase is converted to millimeters after subtracting its mean at the training-selected reference, 141:142/218:219. Pairs lacking a training date or finite/nonzero reference phase are explicitly excluded: 44 are evaluated and none excluded. No spatial offset is fitted. Evaluable locations require finite predictions and valid nonzero observed phase; low-coherence observations are retained. Residuals therefore include held-out observation errors, and evaluability does not establish permanent-gap validity.

All-prediction pooled RMSE is 0.984 mm, versus 0.699/5.150 mm inside/outside the training mask. Training-only support-bin RMSE ranges from 2.338 to 0.412 mm from lowest to highest support; strata use no held-out coherence. On identical evaluable observations, cumulative predictions, linear velocity times pair interval ($365.25 \text{ days yr}^{-1}$), and zero displacement yield RMSE 0.984, 11.751 and 11.784 mm. The velocity comes from training HDF5 with the same final reference; neither baseline fits held-out measurements. All held-out dates remain represented in training, so this tests edge reconstruction, not unseen dates. Re-

constructed atmospheric phase may contribute to the apparent improvement. Shared dates and atmospheric components preclude interpreting these residuals as independent deformation accuracy or pixel/pair counts as effective independent sample sizes. Outputs retain five support thresholds (0.2, 0.3, 0.5, 0.7, 0.8), all four confusion categories, population/area weights, bin counts and quality medians.

Outputs in `results/timeseries/` include `support_finalmask_confusion.csv`, `support_quality_bins.csv`, `heldout_pairs.csv`, `heldout_support_bins.csv`, `heldout_prediction_comparison.csv`, `descriptive_correlations.csv` and `audit.json`. The audit checks twelve stage completions, exclusion of held-out edges, population conservation, pooled residual reconstruction and independent recomputation of selected pair residuals.

Local GNSS feasibility. We additionally screened archived NGL IGS20 daily series for CUSV and CUUT against the 104 full-stack filtered epochs in 2019–2020. Official signed LiCSAR E/N/U coefficients project GNSS into LOS; affine containment selects the InSAR pixel. Exact-date matches number 97 and 66; exploratory ± 7 -day medians increase these to 101 and 77. The stations are approximately 8.6 m apart and occupy the same pixel, 52.2 and 59.8 m from its center. They therefore do not provide two spatially independent checks. GNSS is in a global frame, whereas the inversion is relative to its internal reference area. No independent common datum is established. Containing-pixel and 3×3 -median series, both date windows, source hashes and replay code are retained in the accompanying `Local_Validation_Audit.zip`. First-epoch alignment or separate detrending does not resolve the datum mismatch, so the audit supplies no local accuracy score. Independent validation of the Bangkok inversion requires a common reference frame and spatially independent stations.

S18. Crossed population-model and pair-selection sensitivity

The common domain contains 1,754,190 native GHSL pixels and 22,357,666.843 GHSL population units; 142,949.221 GHSL units from the full primary footprint are excluded. It intersects the primary footprint with full coverage of GHSL 2020 and WorldPop 2020 only; the four-layer epoch comparison in S22 uses a separately defined intersection. Each model is normalized to the GHSL total in this two-model domain before any support calculation. We cross these two allocations with the original 28-pair and expanded 224-pair mean joint support fields, at $q = 0.3$. Neither population product is treated as demographic truth.

All comparisons use 1-km cells in UTM zone 47N and offsets (0, 0), (500, 0), (0, 500) and (500, 500) m. Within each combination, native-model and uniform allocations have identical cell population totals and the same included-domain area. We project extensive area, supported area, population and supported population separately onto a 50-m integration grid. The population–support product is formed before projection. Retaining partial cells avoids full-cell extrapolation outside the domain.

Table S15: Crossed population-model and pair-selection comparison at 1 km and $q = 0.3$. Both models use the same domain and total. Deficits and differences are in millions of allocation units (M). Ranges span four grid origins; they are not confidence intervals. $\Delta = U_{\text{uniform}} - U_{\text{native}}$, and pp denotes percentage points of the common total.

Model	Pairs	Native (M)	Uniform (M)	Δ (M)	Δ/native (%)	Δ/total (pp)
GHSL	28	2.989	4.645–4.653	1.656–1.663	55.38–55.64	7.40–7.44
GHSL	224	3.268	4.944–4.951	1.676–1.683	51.29–51.51	7.50–7.53
WorldPop	28	5.186	5.654–5.657	0.468–0.471	9.03–9.07	2.09–2.10
WorldPop	224	5.488	5.963–5.965	0.474–0.476	8.64–8.68	2.12–2.13

The projected first and cross moments conserve source totals to relative tolerance 10^{-8} , and aggregation over each origin preserves those projected totals. Direct native-grid integration agrees with the coarse cross-moment reference. The 25-m diagnostic retains raw reprojection drift without renormalization (maximum absolute relative moment drift $1.65\text{e-}05$). Its maximum change in uniform-minus-native difference is 0.000436 percentage points of the common total, below the prespecified 0.01-percentage-point numerical tolerance. These are numerical sensitivity checks, not uncertainty intervals.

All 16 combinations have positive uniform-minus-native differences. GHSL gives a larger effect than WorldPop under both pair selections. The result supports a persistent direction for these selected inputs, while showing that a single percentage cannot represent population-model sensitivity. Because pair number and pair composition change together, no causal effect of sample size is inferred. The four origins are deterministic sensitivity cases, not independent replicates.

Complete 16-row results, the 25-m comparison, conservation ledger and input SHA-256 values are retained in `results/cross_sensitivity/`. The analysis and figure can be regenerated with `experiment_cross_sen`

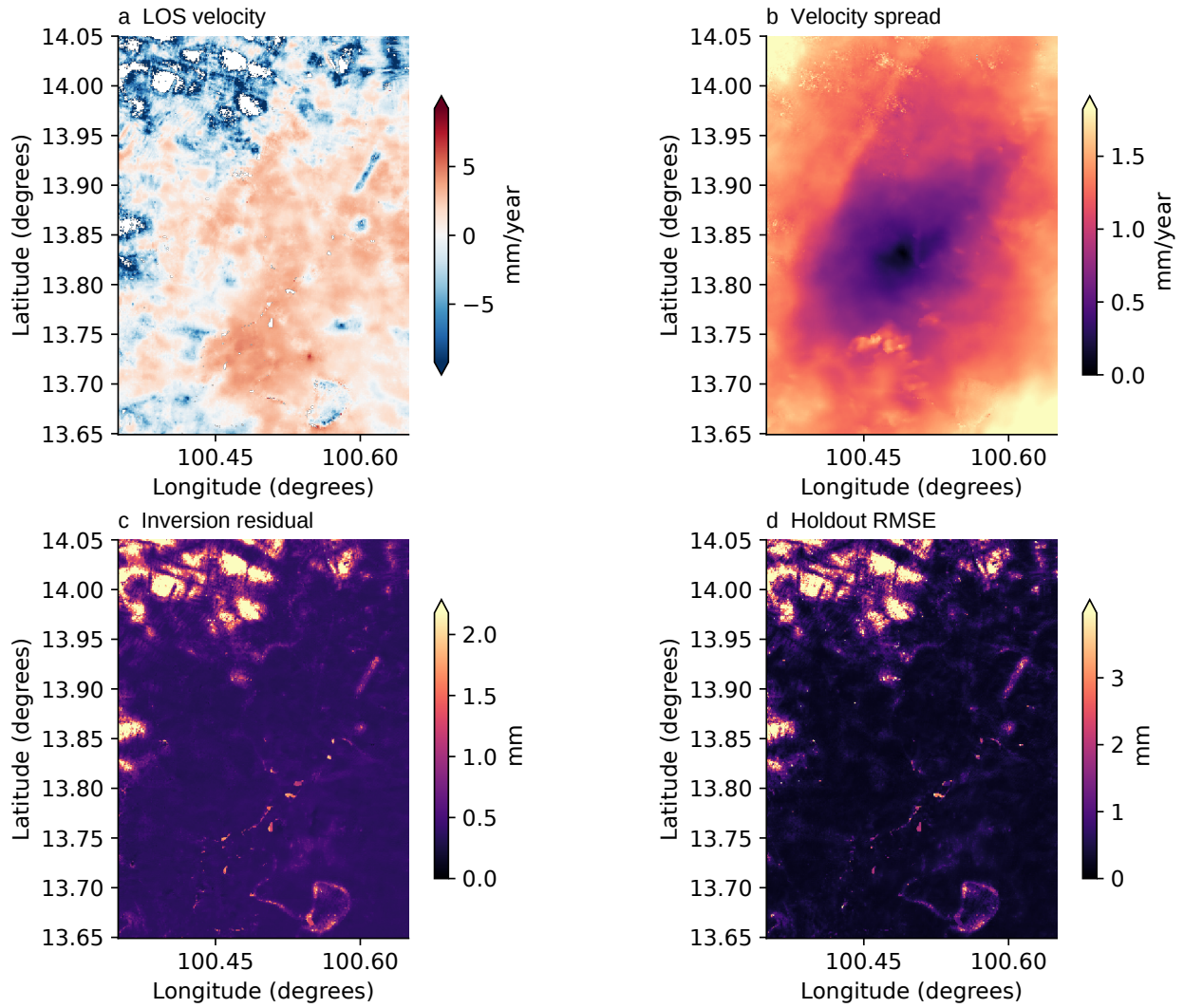


Figure S7: LiCSBAS velocity and quality diagnostics in the Bangkok processing rectangle. Panel a shows the unfiltered fitted LOS velocity only where the full quality mask passes. Panels b–d show bootstrap velocity spread, inversion residual and held-out RMSE, respectively. Symmetric velocity limits and the upper limits of other color scales use their 98th percentiles for readability; values outside those limits are clipped in color only and retained in the data.

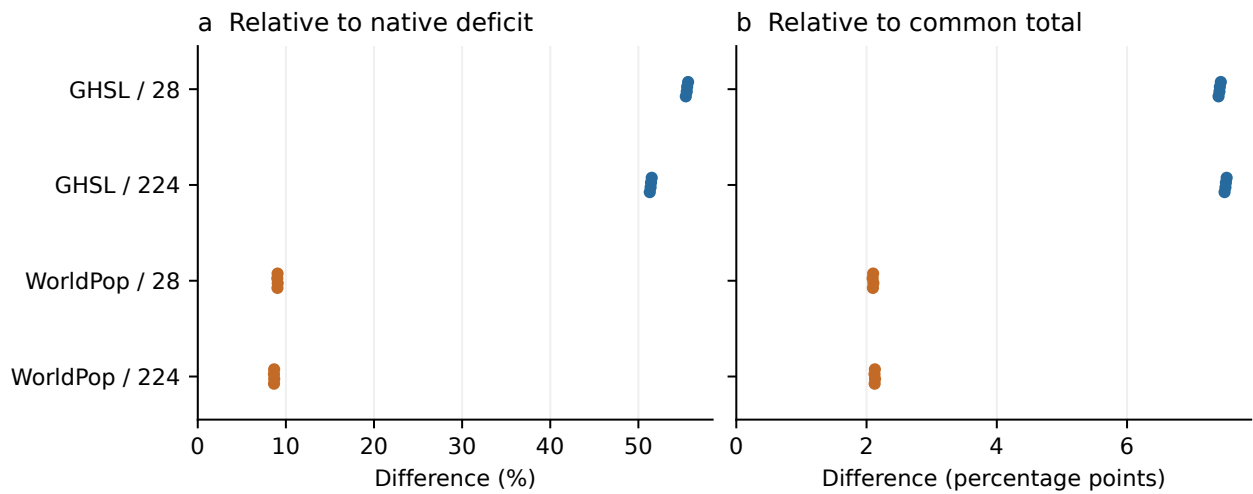


Figure S8: Crossed aggregation sensitivity. Dots show four grid origins and horizontal lines span them. Left: difference relative to the native-model deficit. Right: the same difference as percentage points of the common population total. Colors identify population model; the 28 and 224 labels identify distinct pair selections.

sitivity.py and make_cross_sensitivity_paper.py.

S19. Paired input-support and final-quality-mask allocation

We compared two separate support quantities on the same 401-by-301 LiCSBAS grid and the same fixed GHSL population field: (i) the mean coherence-and-unwrapped support across all 437 input pairs, and (ii) the final full-stack LiCSBAS quality mask. The first is a per-pixel input-pair support fraction; the second is a binary product-level quality decision. They are not interchangeable measurements. For each field we retained population, spherical pixel area, area-times-support and population-times-support separately. We compared native-cell deficits with uniform redistribution within 1-km UTM cells at four zero/half-cell origins.

The direct-geometry audit independently intersected each coarse UTM cell with the original 0.001-degree source cells in longitude-radians/sine-latitude equal-area coordinates. Inverse-projected cell boundaries were densified at 24 segments per edge. This is independent of GDAL raster reprojection and preserves partial edge cells. Uniform-within-source-pixel population and support are assumed during the subpixel intersection. Across all four origins, population mass drift was below 2×10^{-9} allocation units and spherical-area drift was below 10^{-6} m² at printed precision. The direct integration agrees with the refined 25-m raster calculation to the specified 0.01-percentage-point numerical tolerance.

Table S16: One-kilometre allocation effects from direct source-cell intersections. Differences are signed percentage points of the fixed AOI population; positive values indicate a larger deficit under uniform-area allocation than under native population allocation.

Support definition	Native deficit (% population)	Uniform-area effect (pp; four origins)	Interpretation
437-pair mean input support	3.4539	+3.0662 to +3.1891	Input-pair support frequency
Final full-stack quality mask	0.2944	+0.1077 to +0.1170	Product-level accepted/rejected mask

Both definitions retain an allocation effect after coarse aggregation, but the magnitude differs by more than an order of magnitude. This difference is expected because a 437-pair mean support fraction and a final multi-index mask encode different processing information. The difference between these fields does not measure an error in either field. All origin-specific cell intersections and checksums are retained in `direct_geometry_audit/` and `paired_support_pilot/`.

S20. Buffered spatial recalibration and shared-acquisition influence

We tested spatial transfer of total recalibration using buffered holdouts and quantified sampling influence by deleting shared acquisition dates. Recalibration rescales the released GHSL and WorldPop grids without retraining either population model.

For the same 319 complete Fresno block groups as S26, projected polygon-centroid ranks define five west–east or south–north spatial bands. One band is held out at a time. At 0 m, all other groups train the total multiplier. At 1 or 3 km, training polygons touching or within that distance of the held-out polygon union are excluded. Every retained training set has at least 30 groups, and every held-out group is predicted exactly once per design. Calibration factors use training ACS totals only. Model values and geometry remain fixed. We retain 90 fold–model records, every group assignment and 18 full-domain held-out summaries. The 80 ACS variance replicates quantify the survey component of signed fold-total error, conditional on the fixed population grids; they are not population-model error or evidence of source independence.

Table S17: Spatial recalibration against ACS. Ranges span two spatial axes and three buffers, not independent replicates or confidence intervals.

Released grid	Raw WAPE (%)	In-sample (%)	Held-out WAPE (%)	Held-out total error (%)
GHSL	34.89	33.16	33.24–33.85	−0.03–+3.64
WorldPop	33.22	32.29	32.34–32.98	−0.04–+3.78
Census_2020_P1	19.93	19.93	19.94–20.07	−0.27–+0.58

The in-sample total match cannot be interpreted as out-of-area validation. Held-out GHSL and WorldPop WAPE remains above 32%, and the 3-km west–east design has signed total errors of +3.64 and +3.78%,

respectively, compared with +1.39 and +1.43% for the south–north design. Calibration is imperfectly transferable even within this one domain.

For sampling, each of the 225 dates in the 224-pair pool is deleted with all incident edges, not one edge at a time. We recompute summaries under equal-pair weights and under equal weights for the retained 28 year–quarter strata. All strata survive each deletion. The maximum deletion removes seven pairs. The retained graph has 28–34 components, compared with 29 at baseline, and remains separate from the connected 437-pair processing stack. For $q = 0.3$ population weighting, the equal-quarter change spans -0.3856 to $+0.1647$ pp. Full results cover four support definitions and three allocation weightings (2,700 date–support–weight rows).

This is a delete-date influence diagnostic for fixed observed support values. It neither re-solves the velocity inversion nor treats the disconnected support pool as a processing network. Because shared reference, atmospheric structure and acquisition effects remain dependent, we do not form an iid jackknife confidence interval. Source hashes and graph/stratum checks accompany the fold and deletion tables in `additional/spatial_acquisition/`.

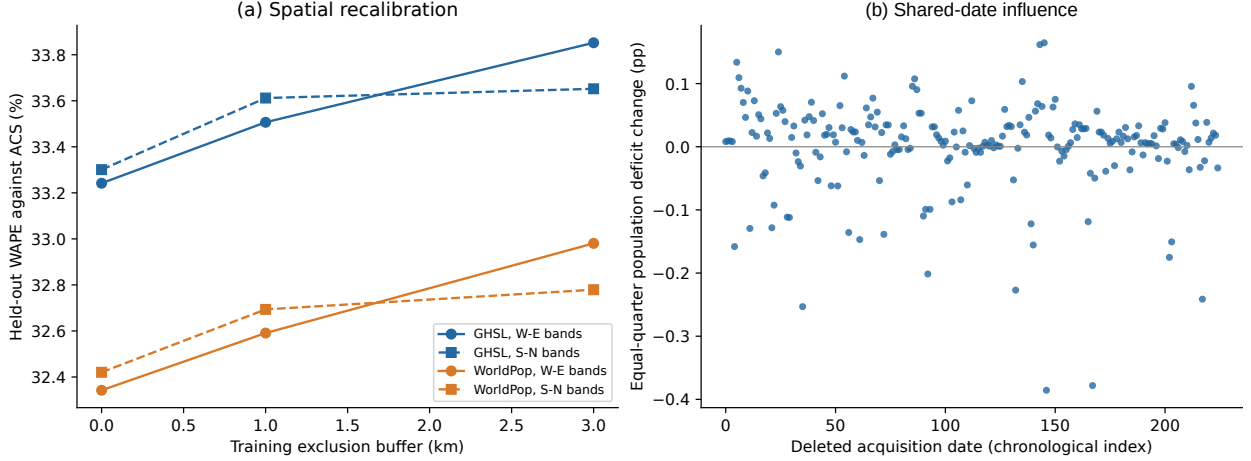


Figure S9: Spatially held-out recalibration and shared-date influence. Left: training-only calibration under two five-band partitions and three exclusion buffers. Right: population-weighted $q = 0.3$ equal-quarter change after deleting every edge incident to one date. Each point is a dependent sensitivity case; no confidence interval or velocity re-inversion is implied.

S21. Training-only quality rules and coverage–residual curves

We specified 18 quality rules before evaluating their coverage and held-out residuals: finite training velocity; the original training mask; training mean-coherence cutoffs 0.05/0.2/0.3/0.4/0.5/0.6/0.7; training residual RMS limits 0.5/1/1.5/2 mm; and training velocity-standard-deviation limits 0.5/1/2/5/100 mm yr⁻¹. Every metric and the reference region are obtained from training pairs only. Rules are post-filters on one fixed training inversion, not repeated inversion fits. No rule is chosen to minimize held-out error.

All 44 excluded unwrapped edges are reconstructed from the retained training time series using their actual dates, the training reference region and the Sentinel-1 phase-to-LOS coefficient. Training and excluded edge sets are disjoint; acquisition dates need not be. Raw recomputation reproduces the frozen S17 finite-velocity pool to a maximum pixel-RMSE difference of 1.69×10^{-7} mm. Each quality rule reports accepted full-domain area/population coverage, evaluable accepted coverage, allocation lacking an excluded-edge residual, pooled pixel-edge RMSE, population-weighted pooled RMSE, pixel median/P95 and per-edge ranges. All accepted population allocation has an evaluable excluded-edge residual in this experiment.

Tightening coherence improves internal residuals here while reducing accepted allocation. Other training metrics need not order the population-weighted error monotonically: the 0.5 and 1 mm yr⁻¹ velocity-standard-deviation filters give population-weighted RMSE 0.288 and 0.360 mm while admitting 4.17 and 31.16% of population allocation. Shared acquisitions, reference and atmospheric signals prevent interpreting these residuals as external velocity accuracy. Full source provenance, 18-rule summary and 792 rule–edge rows are in `additional/training_quality/`.

Table S18: Selected training-only rules. Coverage denominator is the complete matched ROI. Residual units are LOS displacement millimeters; velocity-standard-deviation cutoffs are mm yr^{-1} . All 18 rules and all 44 edge rows per rule remain in the archive.

Rule	Population (%)	Area (%)	Pooled RMSE (mm)	Population RMSE (mm)
Finite velocity	100.00	99.99	0.984	0.359
Training mask	99.86	97.49	0.699	0.350
$q \geq 0.2$	99.40	87.62	0.491	0.344
$q \geq 0.3$	98.27	80.49	0.429	0.340
$q \geq 0.4$	95.34	71.69	0.388	0.336
$q \geq 0.7$	41.04	23.65	0.331	0.317
$\sigma_v \leq 0.5$	4.17	4.11	0.317	0.288
$\sigma_v \leq 1$	31.16	25.55	0.377	0.360

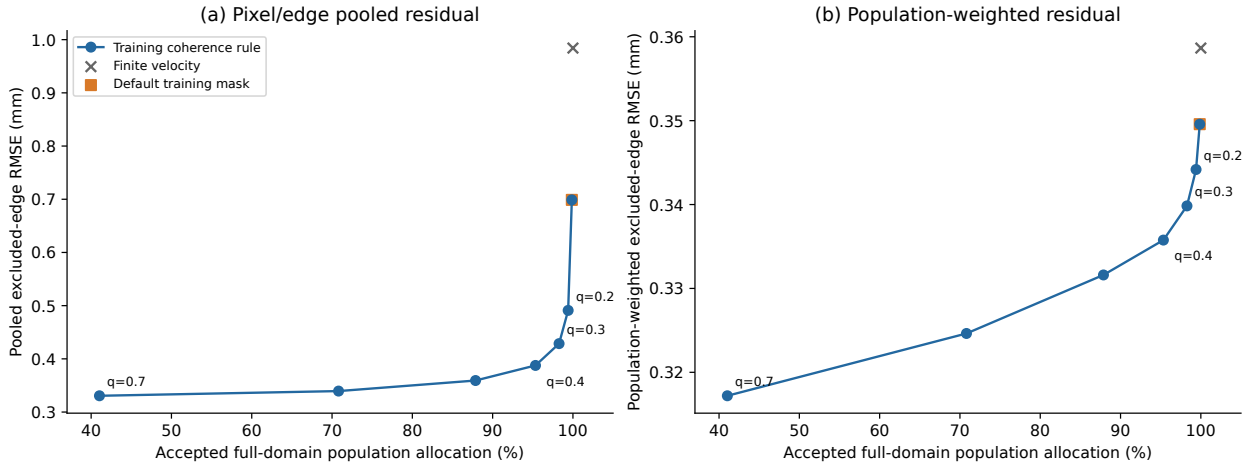


Figure S10: Coverage-residual curve under training-only coherence filtering. Points use the original training mask plus the stated coherence rule; finite velocity and default training-mask baselines are also shown. The right panel weights residual observations by population allocation. Lines connect the coherence-threshold results.

Part E: Population models, administrative benchmarks and residential eligibility

S22. Population coverage and land-cover interpretation

The main population-model comparison restricts to fine pixels covered to within 10^{-5} by all four compared model/epoch rasters. The common domain contains 1,754,190 fine pixels and excludes 142.9 thousand GHSL population units relative to the primary footprint. Outputs on each product's own valid coverage are retained alongside common-coverage outputs. Pixels with NoData are excluded from the common comparison domain rather than assigned zero population.

Normalization multiplies each population model by the same target total divided by its own common-domain total. Its weighted support share is consequently unchanged; this isolates spatial distribution from differences in population totals. The common-total $q0.3$ results are 2.989 million for native GHSL and 5.186 million for WorldPop. Neither establishes demographic truth. A difference between 2015 and 2020 GHSL at fixed 30 arcsec is an epoch/model sensitivity; comparing native 3-arcsecond 2020 to coarse 2015 alone would confound epoch and resolution.

All 11 ESA WorldCover class codes are processed by native fractional intersection. Their fractions sum to one inside the primary domain within 2×10^{-5} . Reported nonzero classes include built-up, tree cover, cropland, grassland, bare/sparse vegetation, permanent water, wetland, shrubland and mangroves. GHSL allocations within a mixed cover pixel remain uniform at the subpixel level. Classification uncertainty, mixed pixels and the 2021 epoch are not converted into a probabilistic population uncertainty estimate.

S23. Fresno 2020 Census-block support overlay

We intersected 2020 Census demographic and housing characteristics block polygons and total-population counts (field P0010001) with the fixed Fresno region of interest used in the DWR analysis. The block service identifies the U.S. Census Bureau DHC tables and TIGER geodatabase as its sources, reports the 2020 vintage, and notes differential-privacy protection of Census counts. We queried 6,999 unique Fresno County block GEOIDs intersecting the ROI; 6,788 complete blocks lie wholly inside its boundary and contain 523,953 Census population, 96.44% of the population in all intersecting queried blocks. The complete-block restriction avoids assigning an entire block population to a partial ROI intersection.

The DWR layer supplies 38,738 published observation polygons, not velocity estimates. We rasterized complete block polygons and the DWR polygon union at fivefold and tenfold supersampling of the 3-arc-second GHSL grid and weighted subpixels by spherical area. For each complete block, the area-uniform Census estimator assigns its population count in proportion to the fraction of block area inside the union of DWR observation polygons. The block-median-observability estimator classifies a block as observed when at least half of its area is covered, then assigns the whole block population to that category. A separate 90% area-coverage threshold is reported as a sensitivity. These are block-level support statistics, not median-velocity classes.

Table S19: Fresno Census-block support estimates. All rates refer to published observation-footprint coverage, not deformation velocity. The complete-block population is 523,953 in both rasterization runs.

Quantity	5×	10×
DWR-observed area within rasterized complete blocks (km ²)	366.903	366.892
Share of complete-block land area covered by DWR polygons (%)	79.761	79.764
Census population-weighted support, uniform within block (%)	99.186	99.192
Census population in blocks with ≥50% observed area (%)	99.406	99.410
Census population in blocks with ≥90% observed area (%)	98.076	98.034
Native GHSL population-weighted support on the same block domain (%)	99.739	99.739
GHSL minus Census-uniform support share (percentage points)	0.553	0.547

At tenfold resolution, 6,784 of 6,788 blocks receive rasterized area; the four blocks without rasterized subpixels have zero Census population. The median ratio of rasterized block area to Census ALAND+AWATER is 0.9999. The fivefold-to-tenfold change is at most 0.042 percentage points across the reported support shares. The native GHSL allocation total in these same complete blocks is 587,868, 12.20% above the 523,953 Census count. The 0.55-point support-share contrast reflects relative spatial weighting and the within-block allocation assumptions. Multiplying every population weight by the same constant leaves a normalized support share unchanged, so the 12.20% total difference does not itself explain this contrast. The total difference and support-share difference are separate diagnostics, neither a calibrated error estimate nor independent population validation.

This support overlay by itself does not reproduce published Census-block median-rate classification: the DWR polygon layer contains no velocities. Section S28 evaluates block-median rate classes using DWR/TRE ALTAMIRA vertical-displacement rasters. Census and GHSL counts may share source inputs; the Census counts are differentially private. The DWR observation polygons span 2015–2026 whereas the population epoch is 2020. Deformation outside the observed footprint is not inferred. The service responses, rasterization script, per-block tables, grid-resolution checks and SHA-256 values are retained in the analysis bundle.

S24. Fresno population-grid comparison with Census block totals

We assessed the coarse spatial allocation of two public 2020 population grids by transferring their 1-km cell counts to the same 6,788 complete 2020 Census blocks used in S23, S28 (523,953 Census residents; 458.669 km² of block land area). GHSL E2020 R2023A at 1 km is provided in the Mollweide equal-area projection. The WorldPop USA 2020 1-km surface is provided at 30-arcsecond geographic resolution; its cell boundaries were densified before reprojection to Mollweide. For both products, each source-cell count was allocated to blocks in proportion to polygon-intersection area. This assumes uniform population within each source cell. We compare each model’s raw total with Census counts and also normalize model predictions to the common Census total to isolate relative allocation shape. A baseline distributes the Census total among blocks in proportion to Census land area.

Both gridded products have substantially lower normalized block-total error than the area-uniform baseline (Table S20). WorldPop is modestly closer on these block-level metrics, but both retain large errors. The results show that the grids carry useful broad-scale allocation information while leaving substantial

uncertainty at sub-kilometre and within-block scales. The comparison evaluates allocation among Census blocks, while locations within each block remain unresolved. Census counts are differentially private, GHSL uses census-derived population inputs, and the exact source overlap between the Census target and the two product vintages has not been resolved. Normalization fixes the Census total, so the comparison evaluates between-block allocation. The official products are GHSL R2023A and WorldPop USA 2020 1-km; the exact dataset records are cited in the References [18, 19].

Table S20: Fresno Census-block comparison of two 1-km population grids and an area-only allocation baseline. Census totals are 523,953 across 6,788 complete blocks. WAPE is the sum of absolute block-level errors divided by the Census total; normalized models are rescaled to the Census total.

Model	Total	Total bias (%)	WAPE (%)	Pearson r	Spearman r
GHSL E2020 raw	579,167	10.54	61.16	0.603	0.584
WorldPop 2020 raw	557,775	6.46	58.04	0.613	0.613
GHSL normalized to Census total	523,953	0.00*	59.37	0.603	0.584
WorldPop normalized to Census total	523,953	0.00*	57.20	0.613	0.613
Uniform over Census block land area	523,953	0.00*	120.88	0.047	0.304

*Zero by construction after normalization; it is not an accuracy result.

Source-cell counts, derived per-block predictions, software script, source checksums and transfer method are retained in the analysis bundle.

S25. Fine-resolution population-allocation sensitivity in Fresno

We compared two 100-m population allocation surfaces against identical 2020 Census block totals and the same DWR observation-support field. Census P1 block counts were joined to 2020 TIGER/Line block geometries for Fresno County. The fixed ROI contains 6,787 fully enclosed blocks and 523,953 residents. The one-block difference from S23–S24 and S28 reflects a different Census-block geometry source; the 6,788-block S23–S24 and S28 benchmark reports the same 523,953-person complete-block total. We first exclude 1,071 zero-population blocks, leaving 5,716 positive-population blocks. Three additional blocks with 45 residents have no GHSL raster mass; we omit them from all methods to define a common domain of 5,713 blocks and 523,908 residents (99.991% of the complete-block total). Both gridded allocation models are normalized within each block to its Census total. The area-uniform comparator distributes each block total uniformly over its land area. The outcome is the population allocation falling outside the DWR polygon-union support after conservative reprojection of the DWR area and supported-area fields to the 100-m CA-POP grid.

The support deficit is 4,417.5 units (0.843% of the common Census total) for area-uniform allocation, 2,016.3 (0.385%) for within-block-normalized GHSL, and 1,268.4 (0.242%) for within-block-normalized CA-POP. Thus the alternative 100-m model changes the deficit by -747.9 units, or -0.143 percentage points of the common population, relative to normalized GHSL. CA-POP gives a lower deficit over the complete-block domain, although the direction differs among tracts: CA-POP yields a higher deficit in 19 of 125 tracts and a lower deficit in 48; the remaining tracts have effectively equal totals. No tract-level significance or random-sample inference is implied. Block-level spatial dependence was summarized with global Moran’s I and binary queen-contiguity weights (shared edges or vertices) on per-block unsupported fractions. Across the 5,713 common blocks, the unsupported-fraction fields have $I = 0.860$ for GHSL and $I = 0.657$ for CA-POP; the CA-POP-minus-GHSL difference field has $I = 0.399$ across 17,083 undirected neighbor pairs (12 isolates) [20]. Moran’s I is reported descriptively for the fixed Fresno domain; no significance test was performed.

CA-POP was produced by dasymetrically allocating 2020 Census block counts using residential parcels and building footprints [21]. Because this input uses the same Census block totals as the target, this is a conditional within-block allocation-model sensitivity analysis, not independent validation of population locations. The sub-block distribution cannot be verified from the block totals. All methods conserve the common block total by construction. Census P1 and TIGER/Line 2020 inputs, the CA-POP total-population raster, input hashes, source URLs and code are listed in the accompanying analysis record. GHSL mass and land area are conserved to relative numerical errors of 0 and 2.4×10^{-16} , respectively, during reprojection.

We also repeated center-sampling and uniform-versus-native comparisons on 250-, 500-, 1,000- and 2,000-m square reporting grids under four grid origins for each 100-m population model. Per-origin results are supplied in `scale_method_sensitivity.csv`; they report signed aggregate differences, summed local absolute error, top-decile ranking overlap and an analytic Cauchy–Schwarz bound on within-tile covariance error. For both models, the bound exceeds the measured local absolute error at every tested scale and origin. Across origins, the uniform-versus-native aggregate difference ranges from 0.284–0.298 percentage points at 250 m

to 1.456–1.622 points at 2,000 m for GHSL, and 0.387–0.395 to 1.577–1.763 points for CA-POP. Top-decile overlap with the native deficit ranking declines with scale, reaching 0.154–0.429 (GHSL) and 0.231–0.500 (CA-POP) at 2,000 m. Center-sampling differences are also origin-sensitive. Both the aggregate difference and the ranking depend on reporting scale and grid placement. The ranges summarize fixed grid origins.

S26. Fresno population-grid comparison with 2017–2021 ACS block-group estimates

We compared the 2020 GHSL and WorldPop 1-km population-count surfaces with the U.S. Census Bureau’s 2017–2021 American Community Survey (ACS) 5-year total-population estimates (Table B01003) on 2020 block-group boundaries [22, 23]. We selected the 319 2020 block groups whose complete geometry lies within the same predeclared Fresno rectangle used for the observation-footprint analysis. The ACS total across these nonoverlapping groups is 493,571, with an exact aggregated 90% margin of error (MOE) of 4,371 calculated from the 80 ACS variance-replicate estimates [24]. The 2020 decennial Census P1 block counts aggregated to these same groups sum to 491,790, a difference of -0.36% from the ACS point estimate and within its aggregate MOE.

To allocate raster counts to block groups, we intersected source pixels with the block-group polygons in an equal-area projection, assuming a uniform population distribution inside each 1-km source pixel. The raster totals are 545,747 for GHSL and 523,349 for WorldPop, respectively $+10.57\%$ and $+6.03\%$ relative to the ACS point estimate. Rank correlation and block-group mean absolute error (MAE) also show substantial local disagreement (Table S21). Rescaling each raster total to the ACS aggregate reduces MAE only to 513 for GHSL and 500 for WorldPop, in model allocation units. This normalization is calibration to the survey total and is reported only to separate aggregate-total differences from within-domain allocation differences.

Table S21: Population-grid counts compared with ACS 2017–2021 block-group estimates for 319 complete Fresno block groups.

Source/model	Total	Difference vs ACS (%)	Spearman ρ	Block-group MAE
2020 Census P1 blocks	491,790	-0.36	0.813	308
GHSL 2020, 1 km	545,747	$+10.57$	0.540	540
WorldPop 2020, 1 km UN-adjusted	523,349	$+6.03$	0.565	514

The table is an external survey benchmark, not ground truth for household locations. The 319 block groups have median area 0.642 km^2 (range $0.104\text{--}28.366 \text{ km}^2$); 256 (80.25%) are smaller than 1 km^2 . They are therefore not uniformly coarser than the nominal 1-km source cells. Allocations to small groups depend directly on the assumed within-source-pixel uniformity, and the overlay cannot validate subpixel or household placement. ACS is a survey estimate rather than an error-free count; the estimate covers 2017–2021, whereas the grids and decennial Census are 2020. We do not know whether the population-model source information is independent of ACS. This benchmark concerns administrative-unit counts within the declared Fresno domain. Per-group estimates, MOEs, raster overlays, the variance-replicate aggregation and input checksums are retained in `acs_population_benchmark/`.

S27. Municipal residential eligibility as a distinct ancillary reference

The public City of Fresno Existing Land Use layer (hosted FeatureServer layer 19) supplies a classification source distinct from the satellite built-surface layer [25]. We obtain all 123,493 features intersecting the fixed rectangle, check unique ObjectIDs and retain only ObjectID, ELU, ELUText and LayerRefreshDate. Strict residential codes are `rl/rml/rm/rmh/rh/rr/rmht/run`; the broader rule additionally admits `raur/rmx/cmx/cce` mixed-use or agricultural-residential classes. Both semantic rules are frozen before the allocation comparison. Unclassified and unmapped locations remain unknown; they are not assigned nonresidential status. The source metadata and refresh dates do not establish a 2020 land-use vintage.

Within complete Census blocks and the common projected 100-m population grid of S25, we intersect municipal geometry with each block-cell piece in EPSG:3310. Seventeen invalid polygons are repaired using `make_valid`; overlapping class fragments are unioned before measuring area, separately for mapped, strict and broader eligibility. All 88,865 pieces satisfy nested eligibility fractions and the $[0,1]$ bounds. Population remains uniform within its source cell. DWR footprint fraction is also the spatially averaged value in each projected 100-m cell; no house-level radar support is inferred.

We first diagnose where the unrestricted block-normalized population falls: residential eligibility, mapped other land use, or unmapped classification. Strict residential eligibility contains 85.22% of mapped GHSL allocation and 89.30% of mapped CA-POP allocation, but unmapped classification represents 32.22 and 31.26% of their complete denominators. The broader definition gives 85.29 and 89.38% within mapped allocation. These agreement fractions are not independent resident counts. Other land-use allocation may reflect mixed use, group quarters, source-cell mixing, classification error or vintage mismatch; those allocations do not establish actual occupancy.

For allocation sensitivity, each rule is normalized to the same complete-block Census count before support summaries are formed. Area-uniform, GHSL, CA-POP, municipal residential-area uniform and GHSL/CA-POP restricted to eligibility are compared on a joint domain where all six have a positive denominator. The strict domain retains 4,879 blocks and 466,999 Census allocation units (89.13% of the complete total); the broader domain retains 4,885 blocks and 467,490 units (89.22%). Both fall below the exploratory 90% coverage threshold. This rule was introduced in the residential-analysis script before its output was examined; it was not in protocol v1 and is not an external reporting standard. It limits the reporting domain, rather than testing statistical significance. This addition to protocol v1 is documented in the protocol-deviation record. The results describe these restricted common domains and are not extrapolated to the full Fresno domain. Excluded masses are 56,954 and 56,463, respectively.

Table S22: Support deficit on the joint common Census-allocation domain. Each row conserves the same block total within each definition; the strict and broader domains differ and cover only 89.13–89.22% of the complete population.

Allocation rule	Strict deficit (%)	Broader deficit (%)
Area uniform	0.21931	0.21908
GHSL	0.09267	0.09257
CA-POP	0.05721	0.05715
Residential area	0.08886	0.08877
GHSL restricted	0.05076	0.05071
CA-POP restricted	0.04136	0.04132

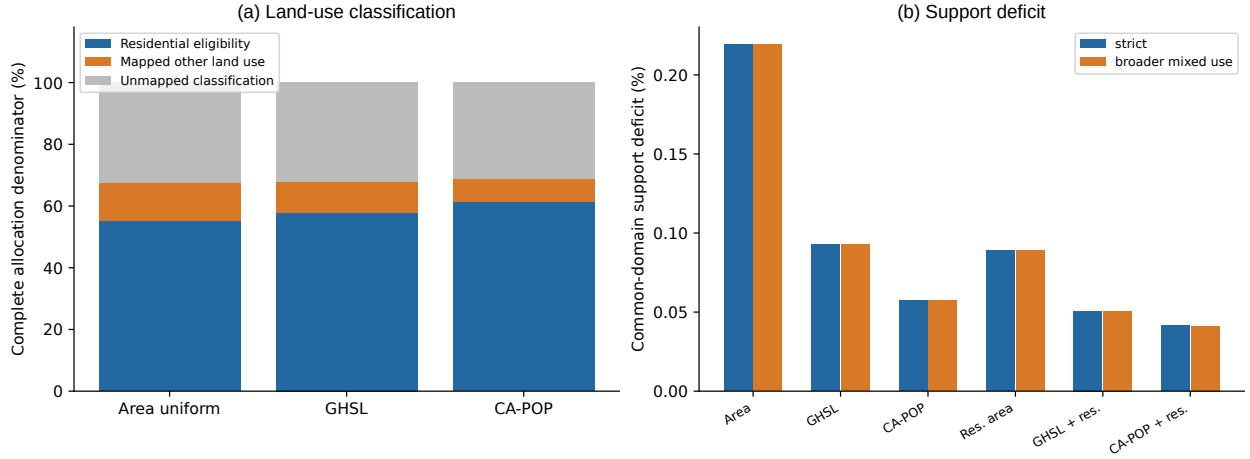


Figure S11: Municipal land-use eligibility and allocation sensitivity. Left: all complete allocation units partitioned into residential, mapped other use and unmapped classification. Right: six rules on their restricted common domain. The restricted domain covers 89.13–89.22% of complete Census allocation. A smaller support deficit after residential restriction does not demonstrate more accurate residential locations.

The municipal layer constrains residential eligibility by land-use class; it does not provide resident counts. Estimates still depend on the GHSL/CA-POP distributions and the land-use vintage. Feature metadata, category definitions, geometry checks, per-block allocations and source hashes are provided in **additional/residential_allocation/** and the selected-source companion.

Part F: Vertical-rate and regional transfer

S28. Census-block median vertical-rate classification in Fresno

To compare with published block-median exposure workflows, we obtained raw floating-point annual-rate and cumulative vertical-displacement rasters from the California DWR TRE ALTAMIRA ImageServer. Although the service is named “Annual Rate Mosaic”, its metadata describe pixel values as vertical displacement in feet over the specified interval. We use its full start/end dates and annualize each raw displacement as $v = 304.8d_{ft}/(\Delta t_{days}/365.2425)$ mm yr⁻¹ before classification or temporal aggregation. The public approximately 100-m rasters are interpolated from point measurements [26, 27]. The six annual rate layers from October 2015–October 2016 through October 2020–October 2021 were exported at their native EPSG:4326 grid size with nearest-neighbor sampling and aspect-ratio adjustment disabled. We built a pixelwise temporal median, retaining only cells valid in all six layers. The temporal-median vertical rate at each DWR cell was then sampled within the published 100-m observation-cell polygons and assigned to Census blocks by polygon membership. The 38,738 observation polygons map exactly to 38,738 native-grid cells in the snapped export. Census-block population is assigned to the median rate within each block, following the transfer step used by Ohenhen et al. [4]. The same convention is used for the cumulative endpoint interval (2015-06-13–2021-01-01) and the irregular-block experiment in S15. The source-window manifest records each service OBJECTID, exact date interval, export request and SHA-256; the six October-to-October windows are consecutive and nonoverlapping. April-to-April results are a distinct seasonal-window sensitivity.

For the six-window temporal median, 6,129 complete Census blocks have valid cells and contain 501,411 people, 95.70% of the 523,953 residents in all 6,788 complete blocks. Blocks with no valid cell are left unclassified rather than assigned to a rate class. The thresholds are $VLM \geq 0$, $-3 \leq VLM < 0$, $-5 \leq VLM < -3$, $-10 \leq VLM < -5$, and $VLM < -10$ mm yr⁻¹, matching the categories in the published comparator. Table S23 reports class-specific populations for the temporal-median and endpoint-rate estimates; Table S24 summarizes annual-rate sensitivity.

Table S23: Fresno 2020 Census population assigned to DWR vertical-rate classes using the median rate in each block. The temporal median uses six October-to-October windows in 2015–2021; the endpoint rate annualizes cumulative displacement from June 2015 to January 2021. Each covers blocks containing 501,411 people.

Temporal estimator	Median-rate class (mm yr ⁻¹)	Population	Share (%)
Six-window median	≥ 0	0	0.00
Six-window median	$[-3, 0)$	42	0.01
Six-window median	$[-5, -3)$	175,524	35.01
Six-window median	$[-10, -5)$	311,231	62.07
Six-window median	< -10	14,614	2.91
Cumulative endpoint	≥ 0	0	0.00
Cumulative endpoint	$[-3, 0)$	197,045	39.30
Cumulative endpoint	$[-5, -3)$	208,738	41.63
Cumulative endpoint	$[-10, -5)$	90,844	18.12
Cumulative endpoint	< -10	4,784	0.95

Table S24: Annual-rate temporal sensitivity. Population percentages refer to Census residents in rate-covered blocks and are assigned by the median annual rate in each block.

Exact interval	Median rate (mm yr ⁻¹)	Coverage (%)	VLM < 0 (%)	VLM < -5 (%)
2015-10-01–2016-10-01	-6.04	95.70	100.00	69.88
2020-10-01–2021-10-01	+7.27	95.70	0.81	0.17
2023-10-01–2024-10-01	-2.66	95.70	92.18	22.92
2025-04-01–2026-04-01	-9.19	95.70	100.00	99.87

For the 2015–2021 temporal-median surface, 64.99% of the rate-covered population falls in blocks with median vertical rates below -5 mm yr⁻¹. Allocating each block population uniformly among its valid DWR cells instead of assigning all residents to the within-block median changes any class share by at most 0.03 percentage points in this period. The annual snapshots vary markedly; for example, the median rate among DWR cells changes from $+7.27$ mm yr⁻¹ in 2020–2021 to -9.19 mm yr⁻¹ in 2025–2026. The cumulative

endpoint mean rate from June 2015 to January 2021 also gives a different block-class distribution from the six-window temporal median. This illustrates temporal-estimand sensitivity and cautions against treating one annual product as a multi-year trend.

To make temporal sensitivity explicit at the block level, we cross-tabulated the median-rate class from the 2015–2021 six-window surface against the 2025–2026 single-year surface using only the same 6,129 blocks with valid rates in both products. We hold the 2020 Census population fixed in both columns to isolate reclassification; the later column is therefore not a contemporaneous 2025–2026 exposure estimate. Of the 501,411 residents in these common blocks, the share assigned to rates below -5 mm yr^{-1} rises from 64.99% to 99.87%, and 61.22% of residents change at least one class. This is a temporal-product sensitivity, not evidence that long-term hazard changed by that amount: the comparison is between a six-year median and one annual estimate. Table S25 reports the full population transition matrix.

Table S25: Fixed 2020 Census population (people) cross-classified by each block’s median DWR vertical rate in 2015–2021 and 2025–2026. Only the 6,129 blocks with valid observations in both periods are included (501,411 residents). The 2025–2026 values are a single-year sensitivity, not a long-term rate or contemporaneous population exposure.

2015–2021 class (mm yr ⁻¹)	2025–2026 class (mm yr ⁻¹)				
	≥ 0	$[-3, 0)$	$[-5, -3)$	$[-10, -5)$	< -10
≥ 0	0	0	0	0	0
$[-3, 0)$	0	0	0	42	0
$[-5, -3)$	0	0	668	174,062	794
$[-10, -5)$	0	0	0	179,154	132,077
< -10	0	0	0	0	14,614

We further tested persistence across the six consecutive October-to-October annual products used in the 2015–2021 temporal median. For each year, we first calculated the median valid vertical rate within each Census block; we then counted how many of the six yearly block medians were below -5 mm yr^{-1} . Among the same 6,129 blocks with valid rates in every window (501,411 residents using fixed 2020 counts), 95.08% of residents are in blocks that cross this threshold in one to five windows, and 0.10% are in blocks below the threshold in all six. Thus 95.17% cross the threshold in at least one window, while 4.83% are never below it. This confirms that single-window classifications are temporally variable in the DWR product; it does not establish which annual rate is correct or imply persistent long-term hazard. Table S26 gives the population distribution by number of years below the threshold.

Table S26: Persistence of the block-median -5 mm yr^{-1} threshold across six consecutive annual DWR products from October 2015 to October 2021. Population uses fixed 2020 Census counts. The denominator is the 6,129 complete blocks with valid DWR cells in every annual window (501,411 residents).

Years below -5	Blocks	2020 Census population	Common population (%)
0	361	24,194	4.83
1	516	36,688	7.32
2	877	61,336	12.23
3	1,335	115,001	22.94
4	1,874	167,391	33.38
5	1,152	96,324	19.21
6	14	477	0.10

The DWR/TRE ALTAMIRA raster is an interpolated vertical-rate product from the same provider as the footprint polygons. The transfer test is limited by interpolation, vintage differences and Census differential privacy; it does not independently validate the original LOS inversion or motion outside the measured footprint. Scripts, cropped rasters, per-block data and checksums are provided in the analysis bundle.

S29. Two fixed additional regions using archived ARIA products

We selected two additional regions before calculating their allocation results: Los Angeles $[-118.4, -118.1]^\circ$ E by $[33.85, 34.10]^\circ$ N and Davis–Sacramento $[-121.9, -121.6]^\circ$ E by $[38.4, 38.7]^\circ$ N. We use GHSL R2023A 2020 population at 3 arcsec and the April 2026 archived ARIA California products of Sangha et al. [28, 29].

Los Angeles uses ascending track A064 (reference P597); Davis–Sacramento uses A137 (P266). Completed notebook outputs and the published track table document periods 14 February 2016–13 May 2022 and 3 March 2017–29 August 2023. Older archived A137 configurations report conflicting dates; the completed notebooks and track table define the periods used here. The GeoTIFF tags alone do not certify a temporal model.

Support here means finite, nonzero published LOS velocity together with mean spatial coherence at least 0.2, 0.3 or 0.4. The zero exclusion follows the provider notebook. This differs from individual-pair unwrapped/coherence support and from a final LiCSBAS mask; identical cutoff values do not make these support definitions equivalent. No physical rate threshold is used for the population-support comparison.

We intersect the published raster support with GHSL source pixels and the fixed rectangles using spherical latitude/longitude area fractions, preserving extensive counts. Integration produces a support fraction for each clipped native GHSL cell; that fraction is held constant within that cell for subsequent reporting-grid comparisons. Thus the larger-scale exact partition applies to this declared piecewise-constant integrated field, not to a household-level support map. The fixed totals are 3,638,746.86 allocation units for Los Angeles and 167,055.45 for Davis–Sacramento. Every 250/500/1,000/2,000-m grid and zero/half-cell origin conserves that total. Signed allocation differences, local L1, Cauchy bounds and top-decile allocation-deficit overlap are retained for all 96 designs. Bounds are checked per reporting cell; relative closure tolerances are below 10^{-9} . Ranking overlap is not a measurement-priority or physical-risk validation.

Table S27: Uniform-minus-native allocation differences (pp of fixed population total) across all four scales and four origins per cutoff.

Region	Coherence cutoff	Minimum (pp)	Maximum (pp)
Los Angeles	0.2	0.0255	0.1567
Los Angeles	0.3	0.0255	0.1567
Los Angeles	0.4	0.0257	0.1859
Davis–Sacramento	0.2	0.0511	0.3723
Davis–Sacramento	0.3	0.0511	0.3723
Davis–Sacramento	0.4	0.0511	0.3723

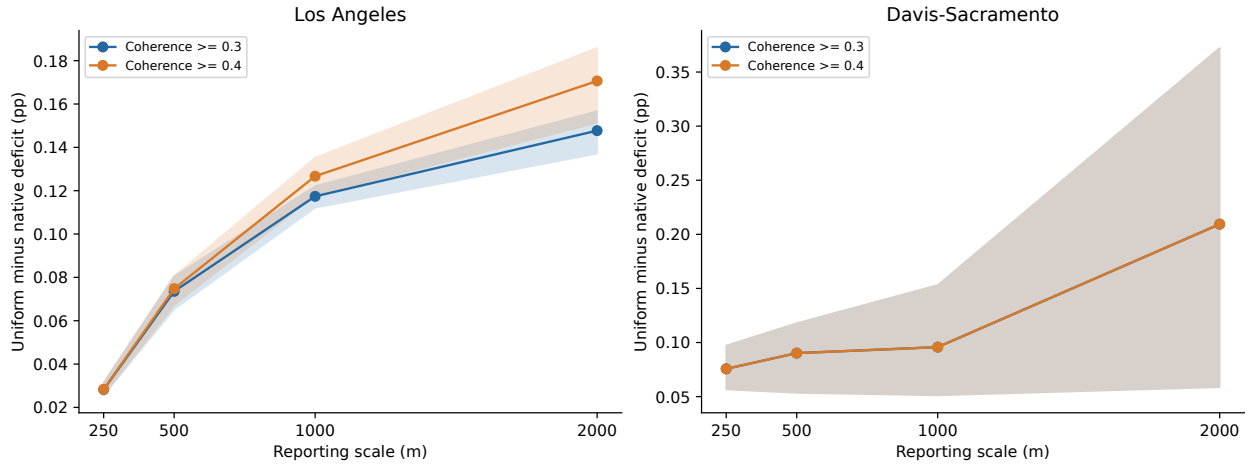


Figure S12: Allocation effects in the two fixed additional regions. Lines are means across four origins; bands span those origins and are not confidence intervals. Cutoffs 0.3 and 0.4 coincide in Davis–Sacramento and partly overlap in Los Angeles. Effects in both regions are smaller than in the primary case.

S30. Archived independent GNSS transfer with disjoint frame controls

We use independent NGL GNSS observations already projected into the LOS geometry and track periods in the completed Sangha et al. notebooks [28, 29]. Their station screening is retained, including the provider completeness and residual-scatter criteria; no station is selected using our comparison residual. Notebook cell 38 reports 411 A064 and 341 A137 LOS rates, explicitly converted from m yr^{-1} to mm yr^{-1} . We join station

names to the public NGL coordinate metadata downloaded on 1 October 2026 [30]. The fixed rectangles contain 12 and five eligible archived stations. The provider reference P597/P266 lies outside the respective rectangle but is retained as a frame control.

Before reading residuals, station coordinates alone select the eligible site closest to each rectangle’s center as an alternative local control: MHMS and UCD1. Provider reference and local-control stations are excluded from every validation metric, including metrics using the other control. Thus the same 11 Los Angeles and four Davis–Sacramento validation stations are compared under both frames. For each control r , both products use the relative quantity $v_i - v_r$; no regression or spatial ramp is fitted to validation stations. The nearest-center choice is an exploratory local-frame sensitivity, not an externally validated optimal reference.

Native samples use the containing cell, calculated with floor through rasterio.index, not rounded pixel-corner indices. Sensitivities use the median within one- and three-pixel-radius windows. Published zero exclusions are preserved except at the provider reference itself. All 19 selected site windows are losslessly archived as 7-by-7-pixel float32 raster subsets, together with the complete source-raster hashes, row/column indices and center coordinates; the full GeoTIFFs are also supplied in the source companion. Reproduction from these raster subsets gives the same station and summary tables as the complete rasters. Maximum station-to-containing-pixel-center offsets are 57.96 and 40.10 m. All sites and every unavailable sample are recorded; none is removed for a large residual.

The exact archived MintPy implementation confirms that `get_los_obs` with an omitted model argument invokes a polynomial degree-1 fit, without periodic or earthquake-step terms. The published InSAR trend uses additional seasonal/step terms. Although periods and spatial frames follow the completed notebooks, the temporal models are therefore not identical. The April data release also retains conflicting older A137 configuration dates. These are source-version and trend-comparison limitations, not presumed corrections. We use the archived linear GNSS fits without refitting the daily series.

Table S28: Relative LOS comparison with archived independent GNSS observations. Bias is at the containing native pixel; RMSE columns give pixel radii 0, 1 and 3. All units are mm yr^{-1} . Controls are excluded from every metric.

Region	Control	n	Bias	RMSE 0	RMSE 1	RMSE 3
Los Angeles	P597	11	-14.43	14.49	14.43	14.34
Los Angeles	MHMS	11	+0.01	1.30	1.32	1.38
Davis–Sacramento	P266	4	+6.68	8.94	9.08	8.98
Davis–Sacramento	UCD1	4	+9.07	10.85	11.16	11.21

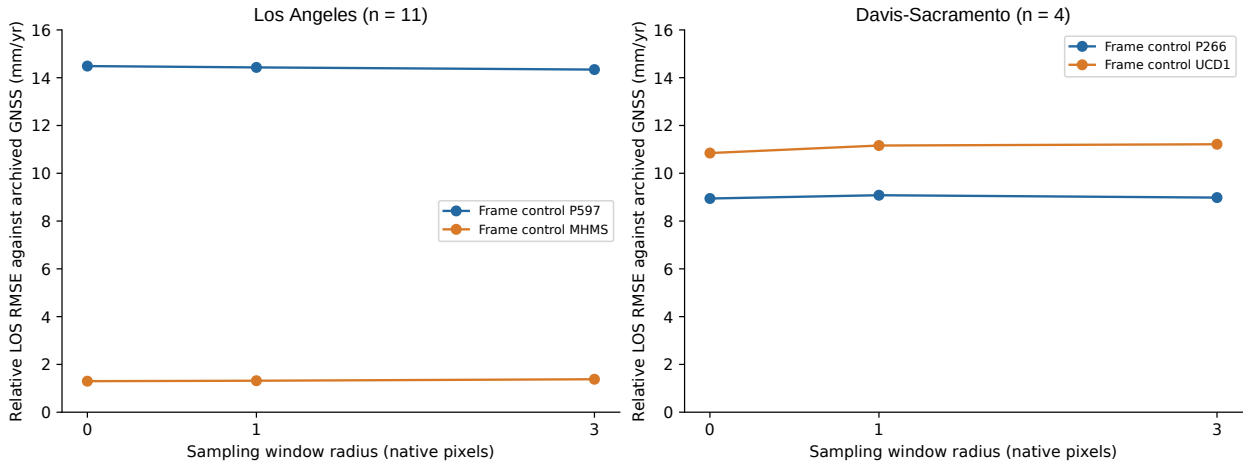


Figure S13: External LOS/GNSS consistency under two disjoint frame controls and three sampling radii. The same validation sites are used within each region. Re-referencing lowers the Los Angeles comparison RMSE but increases it in Davis–Sacramento. These comparisons concern the archived California products and do not validate the Bangkok inversion.

FGNW and FGSW are within 100 m and are assigned one collocation cluster. Los Angeles therefore has 11 validation stations but ten descriptive clusters; Davis–Sacramento has four and four. At radius 0 with local controls, cluster-median RMSE is 1.20 and 10.85 mm yr^{-1} . Shared frame-control error, nearby

monuments and spatial signals prevent treating station errors as independent trials; no iid confidence interval is reported. Radar scattering targets and GNSS monuments may also measure different local motion.

The GNSS comparisons assess the archived California LOS products. The large Davis–Sacramento residuals and differences in temporal models limit conclusions about external accuracy. These comparisons do not test the Bangkok inversion or deformation at unobserved locations. Station-level estimates, all frame/radius summaries, exclusions, collocation assignments, source notebooks and implementation excerpts are provided in `additional/gnss_transfer/` and the source companion.

Part G: Additional figures

S31. Spatial, allocation, masking and processing displays

These displays provide the spatial maps, scale-dependent allocation discrepancies, population sensitivities, controlled-mask errors, matched processing diagnostics and published observation-footprint geometry discussed in the main text. Main Figure 3 gives the continuous support distributions, and main Table 1 gives their integrated deficits. The experiment definitions and complete numerical results are provided in S1–S30.

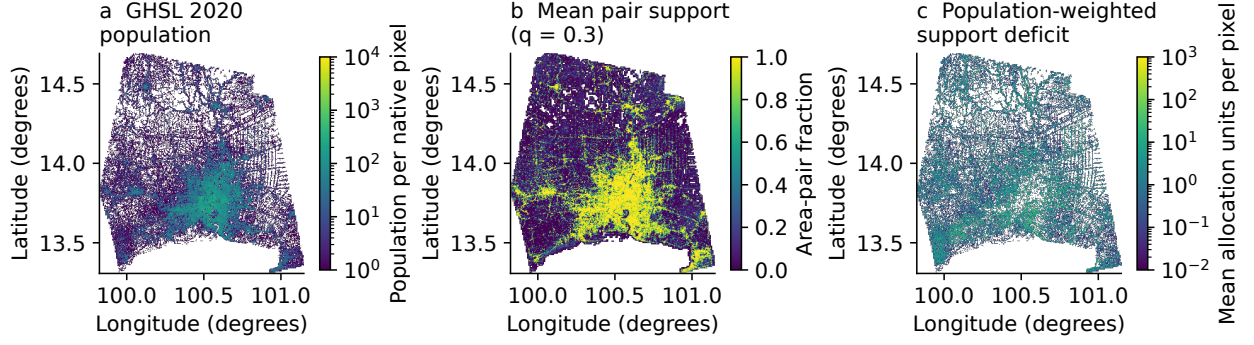


Figure S14: Fixed-footprint Chao Phraya analysis: GHSL 2020 population, mean joint area–pair support at $q = 0.3$, and their allocation-weighted support deficit. Maps use the native GHSL grid; logarithmic population displays leave zero values blank.

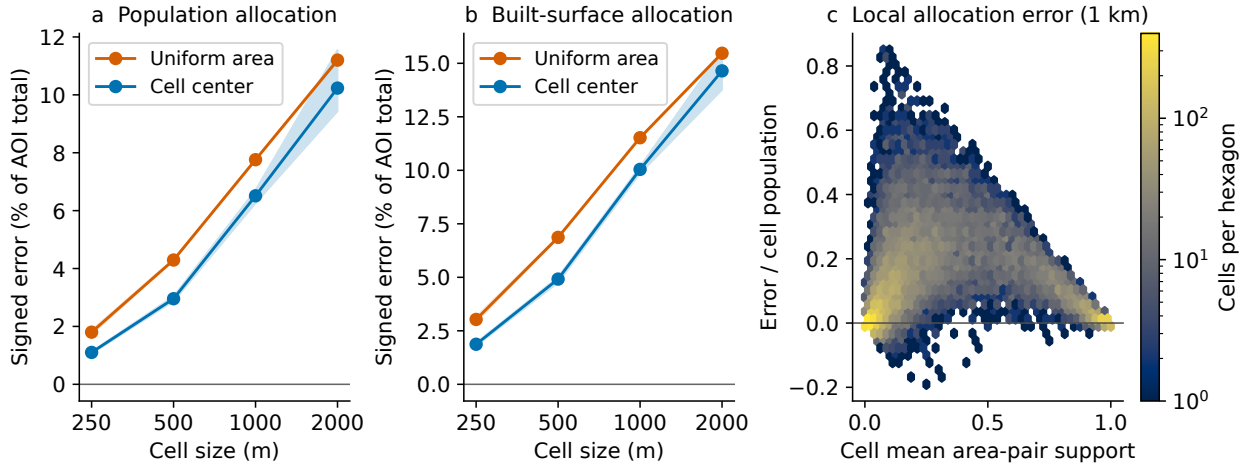


Figure S15: Same-input allocation discrepancies at $q = 0.3$. Lines are means across four origins and bands span those origins, not confidence intervals. Errors are normalized by fixed AOI exposure totals. The local 1-km plot shows uniform-minus-native discrepancy divided by each nonzero cell population; its negative values demonstrate that aggregate overestimation is not universal at the cell scale.

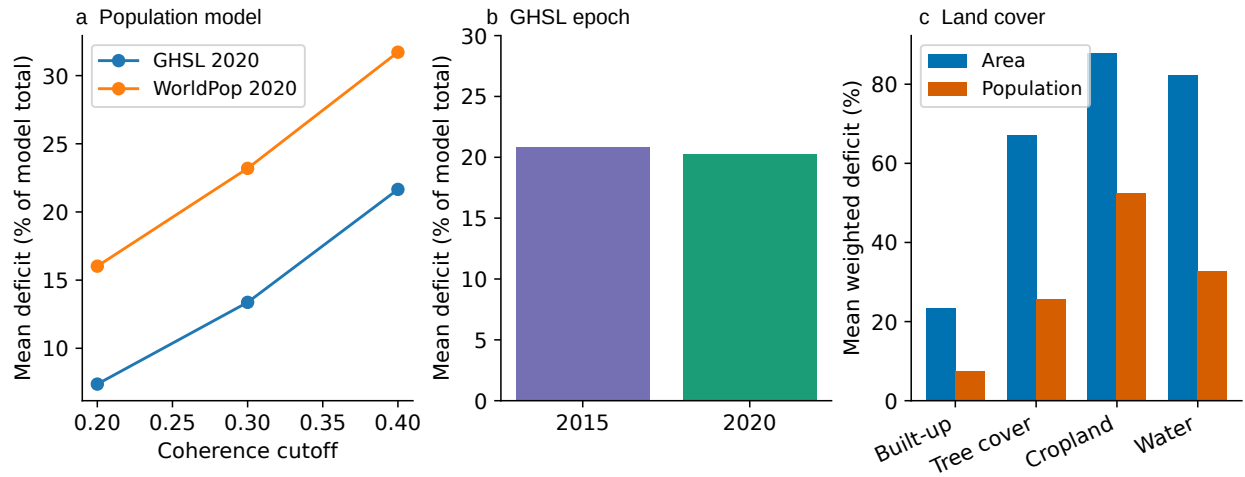


Figure S16: Population and land-cover sensitivities. Population models and epoch comparisons use their common valid domain; the epoch panel uses a fixed 30-arc-second resolution. Fractional land-cover summaries use the primary geographic domain and preserve the within-GHSL-pixel allocation assumption. WorldCover 2021 v200: © ESA WorldCover project 2021; contains modified Copernicus Sentinel data (2021) processed by the ESA WorldCover consortium.

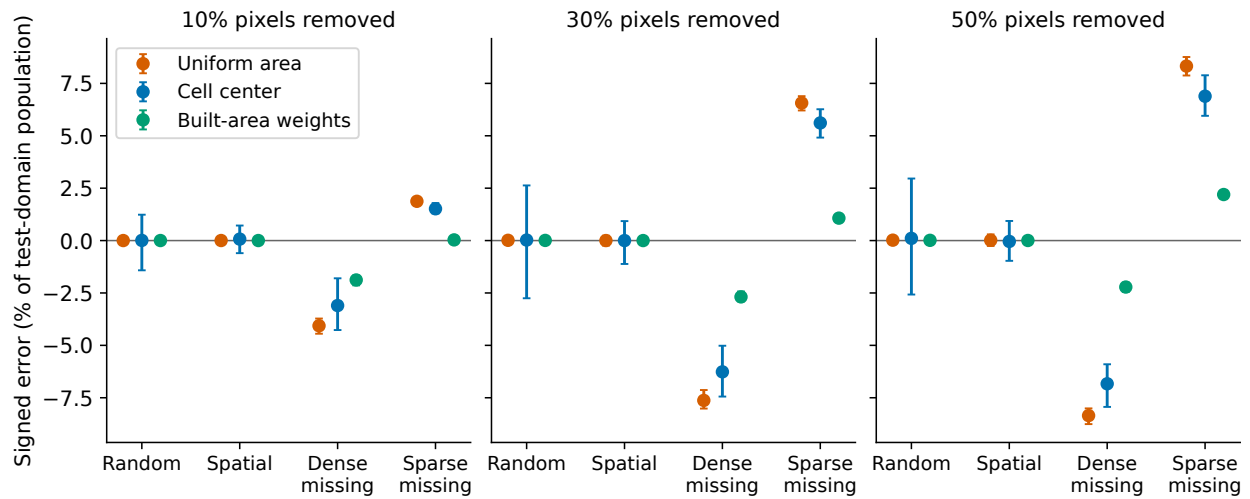


Figure S17: Artificial missingness in the fixed Bangkok rectangle, using 10-fine-pixel coarse cells and zero origin. Symbols show mean signed error and bars the 2.5th–97.5th percentiles of 100 introduced-mask repetitions. The denominator is the entire test-domain population, enabling comparison when the removed population is very small. Complete relative-error, local-error, scale and origin results are provided separately.

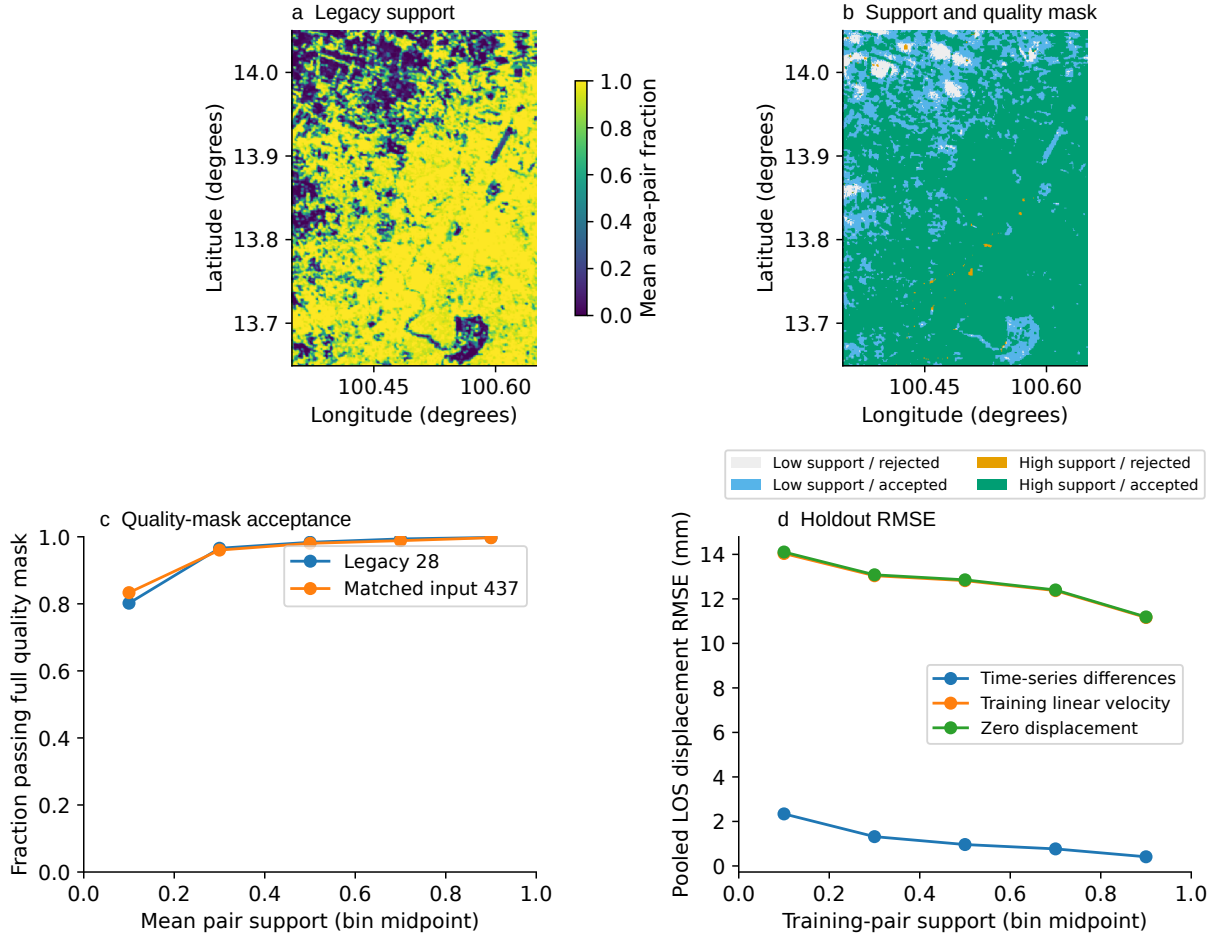


Figure S18: Matched LiCSBAS experiment in the fixed Bangkok rectangle. The legacy support map uses the 2016–2022 sample; its high/low split is 0.5 in panel b, compared with the 2019–2020 full-stack quality mask. Panel c separately includes the matched 437-pair input diagnostic. Both diagnostics use coherence cutoff $q = 0.3$. Panel d groups held-out prediction residuals using training pairs only. Lines connect the support-bin summaries.

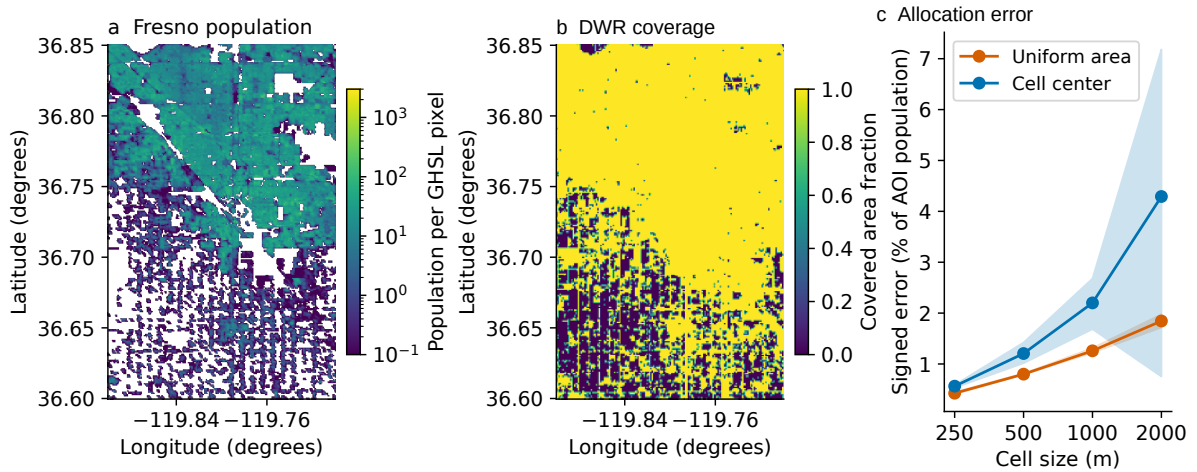


Figure S19: Second-region accounting using actual DWR observation-cell polygons in a fixed Fresno rectangle. Polygon intersections are unioned before assigning population. The deficit reference is small because the observed footprint covers almost all allocated population despite leaving appreciable area uncovered. Scale-comparison errors use the full AOI population as denominator; bands span four orders.

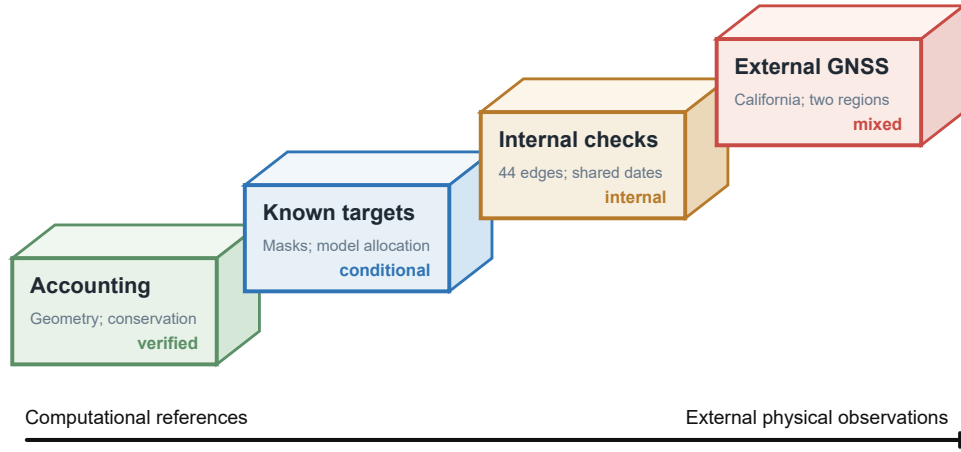


Figure S20: Evidence roles and remaining validation limits. Geometry and conservation test numerical accounting; controlled masks use known conditional references; excluded LiCSBAS edges test internal reconstruction; archived GNSS observations compare an external California product with mixed results. The steps organize distinct evidence targets, not a scale of interchangeable validation strength. Matched independent geodetic validation of the Bangkok inversion and motion truth in permanently unsupported pixels remain absent.

Table S29: Evidence roles and remaining physical-validation limits. Numerical accounting, conditional reference experiments, internal reconstruction and external geodetic consistency address different targets. The external California comparison does not validate the separate Bangkok inversion.

Evidence layer	Completed evidence	Target and limit
Numerical accounting	Geometry, conservation and covariance checks	Correct calculation for the declared inputs; no physical validation
Controlled reference	Artificial missingness; actual-mask class bounds	Error at known reference pixels; bounds on unknown class allocation
Internal reconstruction	44 excluded LiCSBAS pair edges	Reconstruction for dates represented in training; shared-date dependence
Product/region transfer	DWR observation polygons; California allocation cases	Transfer under distinct support definitions; effects are not pooled
Independent observations	Archived GNSS for 11 Los Angeles and four Davis–Sacramento stations	External-product consistency with mixed frame-dependent outcomes; not Bangkok validation
Unresolved physical evidence	No matched independent GNSS/leveling validation for the Bangkok inversion	Deformation in permanently unsupported locations remains unidentified

References

- [1] Blackwell, E., Shirzaei, M., Ojha, C. & Werth, S. Tracking California’s sinking coast from space: Implications for relative sea-level rise. *Science Advances* **6**, eaba4551 (2020).
- [2] Fernandez-Torres, E. A. *et al.* Country-scale assessment of urban areas, population, and households exposed to land subsidence using Sentinel-1 InSAR, and GPS time series. *Natural Hazards* **120**, 1577–1601 (2024).
- [3] Ao, Z., Hu, X., Tao, S., Hu, X. *et al.* A national-scale assessment of land subsidence in China’s major cities. *Science* **384**, 301–306 (2024).
- [4] Ohenhen, L. O., Zhai, G., Lucy, J. *et al.* Land subsidence risk to infrastructure in US metropolises. *Nature Cities* **2**, 543–554 (2025).
- [5] Murray, K., Barbee, M., Thompson, P. & Fletcher, C. Coastal land subsidence accelerates timelines for future flood exposure in Hawai’i. *Communications Earth & Environment* **6**, 123 (2025).
- [6] Aieorattanawadee, N. *Monitoring land subsidence in Bangkok during 2018–2022 using time-series InSAR analysis with MintPy*. Master’s thesis, Chulalongkorn University (2022). URL <https://digital.car.chula.ac.th/chulaetd/6585/>.
- [7] Luachapichatikul, S. *Measuring Land Subsidence with PS-InSAR in Bangkok using Sentinel-1 Time-Series Techniques*. Master’s thesis, Burapha University (2020). URL <https://ir.buu.ac.th/dspace/handle/1513/311>.
- [8] Pumpuang, A. & Aobpaet, A. Evolution pattern of land subsidence using InSAR time-series analysis in Bangkok, Bangkok, Thailand. *Journal of Current Science and Technology* **14**, 49 (2024). URL <https://ph04.tci-thaijo.org/index.php/JCST/article/view/2549>.
- [9] Keerinarat, C., Phumpuang, A., Ladavadee, A., Supavetch, S. & Aobpaet, A. Subsidence rate analyzing of Bangkok soil profile with InSAR data. The 25th National Convention on Civil Engineering, GTE44-1–GTE44-6, Chonburi, Thailand (2020). URL <https://www.conference.thaince.org/index.php/ncce25/article/download/670/265/>.
- [10] Haghshenas Haghighi, M. & Motagh, M. Uncovering the impacts of depleting aquifers: A remote sensing analysis of land subsidence in Iran. *Science Advances* **10**, eadk3039 (2024).
- [11] Payne, J. A. *et al.* Widespread extent of irrecoverable aquifer depletion revealed by country-wide analysis of land surface subsidence hazard in Iran. *Journal of Geophysical Research: Solid Earth* **130**, e2024JB030367 (2025).
- [12] Payne, J. Accompanying data for paper: Widespread extent of irrecoverable aquifer depletion revealed by country-wide analysis of land surface subsidence hazard in Iran. <https://doi.org/10.5281/zenodo.13754200> (2024).
- [13] Morishita, Y. *et al.* LiCSBAS: An open-source InSAR time series analysis package integrated with the LiCSAR automated Sentinel-1 InSAR processor. *Remote Sensing* **12**, 424 (2020).
- [14] Japan International Cooperation Agency. The kingdom of Thailand: Project for capacity development and promotion of utilization of national CORS data center, project completion report (2nd and 3rd terms). Japan International Cooperation Agency report (2024). URL https://openjicareport.jica.go.jp/pdf/12386280_01.pdf. SBKK station coordinate in Table 5.
- [15] Eicher, C. L. & Brewer, C. A. Dasymetric mapping and areal interpolation: Implementation and evaluation. *Cartography and Geographic Information Science* **28**, 125–138 (2001).
- [16] Mennis, J. Generating surface models of population using dasymetric mapping. *The Professional Geographer* **55**, 31–42 (2003).
- [17] Yunjun, Z., Fattahi, H. & Amelung, F. Small baseline InSAR time series analysis: Unwrapping error correction and noise reduction. *Computers & Geosciences* **133**, 104331 (2019).
- [18] Schiavina, M., Freire, S., Carioli, A. & MacManus, K. GHS-POP R2023A: GHS population grid multitemporal (1975–2030). <https://doi.org/10.2905/2FF68A52-5B5B-4A22-8F40-C41DA8332CFE> (2023). URL https://human-settlement.emergency.copernicus.eu/ghs_pop2023.php. Dataset DOI: 10.2905/2FF68A52-5B5B-4A22-8F40-C41DA8332CFE.
- [19] WorldPop and Center for International Earth Science Information Network (CIESIN), Columbia University. Global high resolution population denominators project: United States 2020 UN-adjusted 1-km population grid. <https://doi.org/10.5258/SOTON/WP00671> (2018). URL <https://hub.worldpop.org/geodata/summary?id=34816>. Dataset DOI: 10.5258/SOTON/WP00671; production date 2020-06-22.
- [20] Moran, P. A. P. Notes on continuous stochastic phenomena. *Biometrika* **37**, 17–23 (1950).
- [21] Depsky, N. J., Cushing, L. & Morello-Frosch, R. High-resolution gridded estimates of population sociodemographics from the 2020 census in California. *PLOS ONE* **17**, e0270746 (2022).
- [22] U.S. Census Bureau. 2021 american community survey 5-year summary file (2021). URL <https://www.census.gov/programs-surveys/acs/data/summary-file/sequence-based/2021.html>.
- [23] U.S. Census Bureau. 2020 TIGER/line shapefiles: Block groups (2020). URL <https://www2.census.gov/geo/tiger/TIGER2020/BG/>.
- [24] U.S. Census Bureau. 2017–2021 ACS variance replicate estimate tables documentation (2022). URL https://www2.census.gov/programs-surveys/acs/replicate_estimates/2021/documentation/5-year/2017-2021_Variance_Replicate_Table_Documentation.pdf.
- [25] City of Fresno. Public existing land use GIS layer (2026). URL https://services2.arcgis.com/WkBUoUjyNPhsW0k1W/arcgis/rest/services/Existing_Land_Use/FeatureServer/19. Hosted FeatureServer layer 19; ObjectID, ELU, ELUText and LayerRefreshDate; accessed 1 October 2026.
- [26] California Department of Water Resources. Vertical displacement point data locations, 2026q1: published

- observation-cell polygons (2026). URL https://gis.water.ca.gov/arcgis/rest/services/Elevation/Vertical_Displacement_Point_Data_Locations_2026Q1/MapServer. Service layer 0, geometry type polygon; accessed 27 September 2026.
- [27] California Department of Water Resources. SGMA data viewer documentation: TRE Altamira InSAR dataset (2026). URL <https://sgma.water.ca.gov/webgis/config/custom/html/SGMADataViewer/doc/>. 100 m measurement support distinguished from interpolated display rasters; accessed 27 September 2026.
 - [28] Sangha, S. S., Funning, G. J., Govorcin, M. & Bekaert, D. P. S. A statewide InSAR velocity map for California produced by a comprehensive large scale analysis of ARIA standard products. *Earth and Space Science* **13**, e2026EA005214 (2026).
 - [29] Sangha, S., Funning, G., Govorcin, M. & Bekaert, D. Data and software for sangha et al., 2026: A statewide InSAR velocity map for California. <https://doi.org/10.5281/zenodo.19493073> (2026). Fixed April 2026 archived version; CC BY 4.0.
 - [30] Nevada Geodetic Laboratory. GNSS station data holdings and coordinate metadata (2026). URL <https://geodesy.unr.edu/NGLStationPages/DataHoldings.txt>. University of Nevada, Reno; accessed 1 October 2026.